DYNAMIC: Rain/Wetness-Manual

Company: MoveRotateResize
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Feb. 05th, 2025
Version# 1.3.1

Demo
Youtube
Link Tree
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Overview

Effortlessly add Rain/Wetness to all your game objects! Built with Shader Graph to allow complete user freedom and customization.Includes over 11 Shaders,Rain Occlusion,VFX,Rain Audio,Terrain Puddles

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Compatible Unity Versions

1. 2022.2 +
2. 2022.3 +
3. 2023.2 +
4. 2023.3 + Alpha/Beta
5. Unity 6

Compatible Render Pipelines

- HDRP
- HDRP- Raytracing



Included in Package

Shaders

Transform your Unity projects with the Dynamic RainWetness Shader, a cutting-edge shader crafted with Shader Graph. This powerful tool enhances the visual fidelity of your materials, delivering detailed and immersive experiences. Thanks to Shader Graph, customization and modification are straightforward, allowing you to tailor the shader to your specific needs effortlessly.

Shader List

Rain/Wetness: Full featured shaders with RainWetness integration. • Default • Translucent • Glass • Subsurface • Anisotropy • Iridescence	***Rain/Wetness- Tessellation:*** Full featured shaders with RainWetness integration. Tessellation Currently Only Makes Mesh Appear higher poly • Default Tessellation • Translucent Tessellation • Glass Tessellation • SubsurfaceTessellation • Anisotropy Tessellation • Iridescence Tessellation

Key Features:

- Rain/Wetness Support
- Detail Map Support
- Mask Map Support
- Tangent/Object Space Normal Support

Rain/Wetness Masks

Global Rain (Global Variable):

- A glo

Rain/Wetness Settings

Adds rain effects like wetness, rain streaks, and raindrops that will dynamically cover your game objects. Rain streaks and drops are dynamically placed on game objects based on gravity like settings. In other words this means that the raindrops appear on the top and sides of objects, and rain streaks appear only on the sides of objects. You can also control the porousness of game objects; which lets you specify if a game object is nonporous or porous. The latter will result in the game object appearing more wet with brighter colours. As an example, think of how rocks look when it's raining; they appear more wet and their colour gets more saturated. This effect can be controlled via sliders, Enviro- Sky and Weather Integration, or via the Dynamic Weather Rain script.

Global Rain (Global Variable):

- A global shader variable that controls the amount of global rain coverage
- This variable is controlled by the DYNAMIC: Rain Wetness scrip

Rain Occlusion:

★ **Don’t touch this variable as it’s used by the occlusion system**

- Controls the rain coverage per gameobject. This is used by the DYNAMIC: Occlusion and Occlusion Zone scripts, as they work together to dynamically add/remove rain from gameobjects as they enter/exist covered areas (Occlusion Zones)

Smoothness Override:

- Lets you adjust how wet your gameobject appears when it’s raining. This override the smoothness value by adding this value on top of it

Normal Override:

- Lets you adjust the normal strength of your gameobject when it's raining. This override the normal value by adding this value on top of it

Porous/Non-Porous Slider:

- Porous Material (wood, drapes, carpet, and cardboard are porous)
    - Saturates colours
    - Darkens colours
- Non-Porous Material (glass, metal, plastic, and varnished wood)
    - Saturates colours
    - Makes objects look wet

Rain Drop Texture: ● R= Normal X ● G= Normal Y ● B= Fade Mask ● A= Stationary Rain Drop Mask	***Rain Drop Settings:*** ● Speed (Global Variable) ● Normal Strength ● Smoothness Strength

Rain Streak Texture: ● R= Normal X ● G= Normal Y ● B= Rain Streak Mask ● A= *Not Used	***Rain Streak Settings:*** ● Speed (Global Variable) ● Distortion Size (Global Variable) ● Distortion Normal Strength ● Normal Strength ● Smoothness Strength

VFX Shaders

Transform your Unity scenes with our advanced weather effects, designed to bring unparalleled realism to your environments. Using Unity's VFX Graph or Particle System, you can simulate everything from light showers to heavy storms, adding dynamic rainfall to your scenes. Enhance your terrains with reflective puddles, glossy wet surfaces, and atmospheric ground fog. Our fog masks are specially designed to mask out fog in interior spaces, ensuring a seamless blend between outdoor weather effects and indoor environments. Perfect for games and simulations, these weather effects create stunning, immersive worlds with unmatched depth and realism.

Shader List

Fog:

- Fog- Ground (Shadergraph)
- Fog- Mask (Shadergraph)

Falling Rain:

- Falling Rain (VFX Graph) with...
    - Depth Buffer Occlusion
    - VFX Occlusion Zones
    - SDF Occlusion
    - Aerial Camera Occlusion

- Falling Rain (Particle System) with...
    - Gameobject Mesh Collider Occlusion

Dripping Rain:

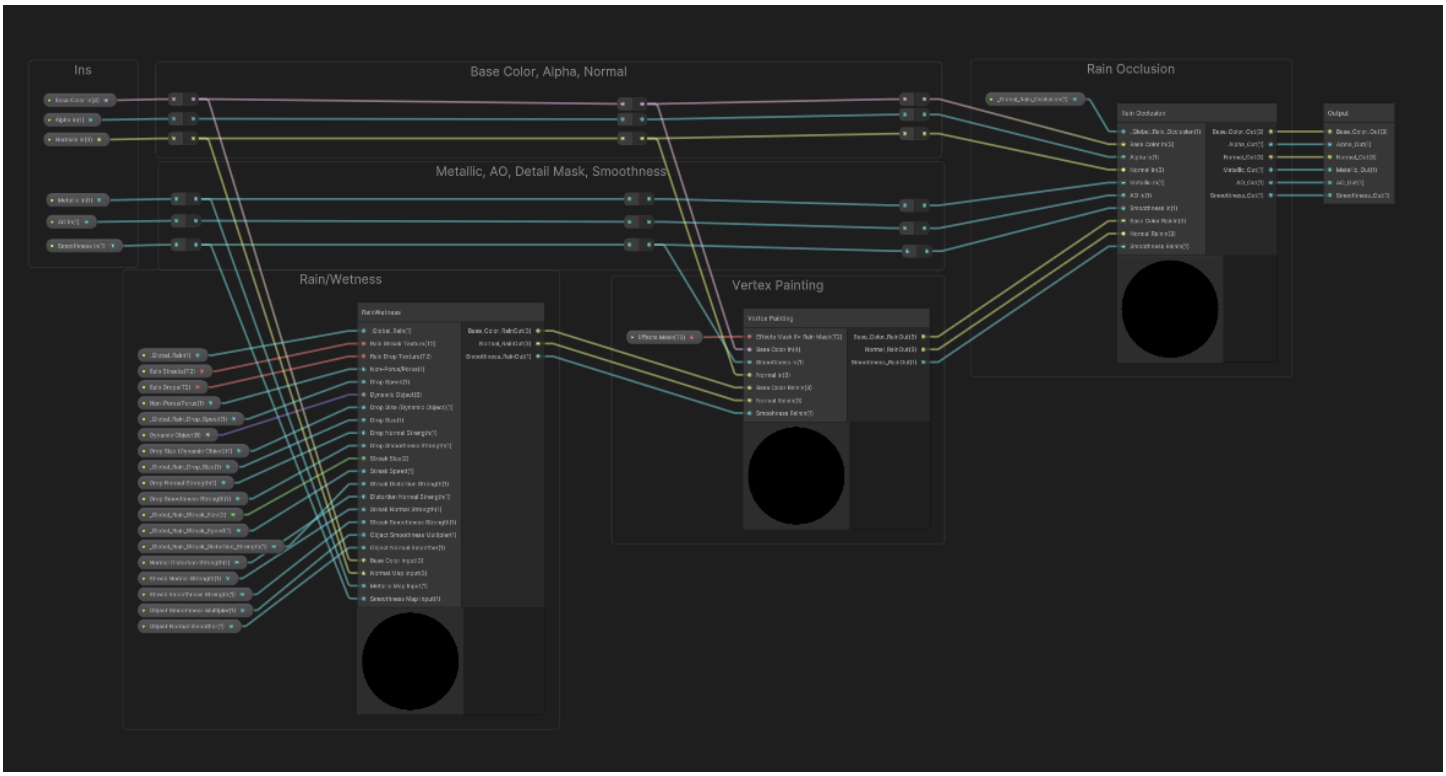
- Dripping Rain- Static (VFX Graph)
- Dripping Rain- Moveable (VFX Graph)

Terrain Decals:

- Terrain Puddles (Shadergraph)
- Terrain Wetness (Shadergraph)

Shader Subgraph- Rain/Wetness Systems

Integrate the full suite of Rain/Wetness features into your own Shader Graph shaders! Easily add all of the advanced rain effects to your gameobjects and take your project to the next level!



Key Features

Seamless Integration:

- Easily add all of the Rain/Wetness features to your own Shader Graph shaders.

Global Rain/Wetness Coverage:

- Easily add rain/wetness to your pre existing shader graph shaders

Rain Occlusion:

- When used with the Dynamic- Occlusion and Occlusion Zone scripts make it so movable gameobjects can dynamically be covered with rain/wetness when moving in and out of Occlusion Zones.

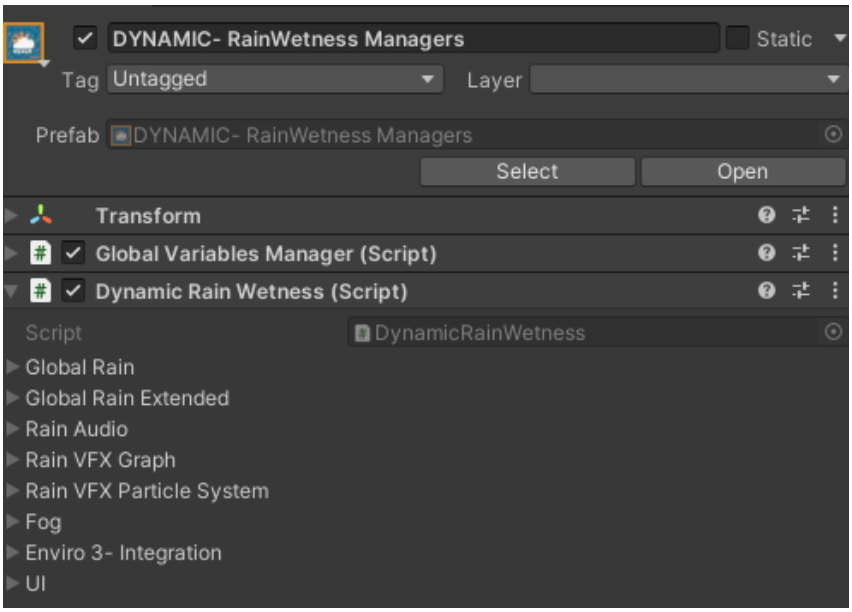
Rain Occlusion via Vertex Painting:

- Allows you to mask out rain/wetness effects. For example paint the interior of a building to mask out the rain effects while having the outside be able to be covered by rain

DYNAMIC: Rain Wetness

The ultimate tool for adding unparalleled realism and immersion to your Unity projects. With full control over rain/wetness effects, this asset allows your environments to react dynamically to changing weather conditions, breathing life into your scenes like never before. Whether you're crafting a light drizzle or a torrential downpour, DYNAMIC-RainWetness gives you complete control to create the perfect atmosphere for any scenario. From lingering puddles to wet surfaces that stay after the rain stops, this tool ensures your world feels alive and immersive.

Key Features



Global Rain Coverage:

- Seamlessly adjust the rain coverage across all objects in your scene. From a light drizzle to a torrential downpour, control it all with a single variable.

Global Rain Transition Duration:

- Enable smooth, realistic transitions for rain effects. Whether you're starting or stopping the rain, this feature lets weather systems apply or remove rain effects gradually, enhancing the realism of your scenes.

Global Rain Extended:

- Extend the life of your rain/wetness effects beyond the rain itself. Perfect for creating lingering puddles, and dripping rain that stay around even after the rain has stopped.

Global Rain Extended Transition Duration:

- Prolongs the transition of some rain effects. Featuring puddles that stick around and disappear with time, and dripping rain from the edges of objects that slows naturally overtime.

Rain Drop Settings:

- **Size:** Adjust the raindrop size to create fine mist-like drops or large, dramatic droplets.
- **Speed:** Control how quickly raindrops appear and disappear for dynamic or calm rainfall.

Rain Streak Settings:

- **Size:** Customize the width and length of streaks on surfaces to match light drizzles or heavy showers.
- **Speed:** Set how fast streaks slide down toward the ground for a realistic rainfall effect.
- **Distortion:** Add dynamic distortion to rain streaks to simulate natural interactions with uneven surfaces.

Main Rain Audio Source:

- Adjusts the volume levels of the Main Rain audio source dynamically when entering/existing buildings, caves, ect. Perfect for adding that extra level of immersion and atmosphere.

Fog Control:

- Manage the intensity of fog for both global and local fog volumes, adding depth and mood to your scenes.

Enviro 3- Integration:

- Enable this to have our systems work seamlessly with Enviro 3- Sky and Weather. This lets you have all of our rain/wetness effects and systems alongside the power of Enviro 3's weather systems.

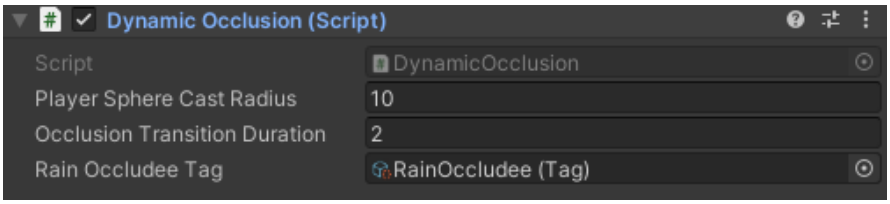
DYNAMIC: Occlusion and Occlusion Zones

A powerful set of scripts that provide dynamic rain/wetness coverage for all movable game objects, including characters, vehicles, items, and more. This tool ensures that rain behaves realistically, adapting to your environment, gameplay, NPC, and player. As the player moves in and out of structures, DYNAMIC-Occlusion also dynamically adjusts the rain audio, enhancing atmosphere and immersion. Additionally, it features Falling Rain VFX Occlusion, ensuring that Falling Rain VFX never passes through buildings or structures, creating a seamless and natural experience.

Key Features

Dynamic Rain/Wetness Coverage Per Gameobject:

- Automatically apply/ remove rain and wetness effects as game objects move in and out of occlusion zones. Perfect for creating realistic interactions with rain, whether it's a character walking into a building or a vehicle driving under a bridge.



Falling Rain Occlusion Zones:

- Our advanced VFX Occlusion system delivers seamless transitions between indoor and outdoor environments. Making sure that rain never enters your buildings, structures, overhangs, ect.

Falling Rain SDF Occlusion Zones:

- Use the power of Unity and VFX Graph to create SDF textures which allow you to mask out rain for complex structures and shapes. Our system includes the ability to combine all of your gameobjects into 1, which allows our rain to be occluded by a single SDF Occlusion Zone. You can also expand upon this if your project calls for it!

Adaptive Interior Rain Audio:

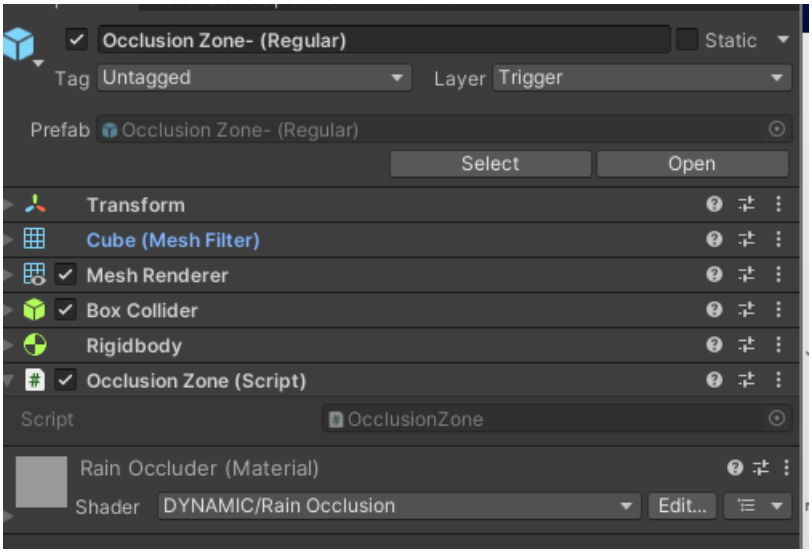
- Controls all rain audio levels, simulating the sound of rain getting louder or quieter as objects move inside or outside of buildings and structures. This feature adds an extra layer of realism, enhancing the overall auditory experience

Edit Mode Occlusion:

- With a single press you can have occlusion be active in the editor. Perfect for visualizing what occlusion will look like when creating your environments . As playmode uses an optimized trigger system to detect occlusion zones and isn't compatible in editor.

Easy Set Up:

- Simply place a few Occlusion Zones in a building. Then select all of your movable gameobjects and add the Dynamic Multi Tag component and set the tag to Rain Occludee and voilà you have rain occlusion for rain audio and your all gameobjects that use a



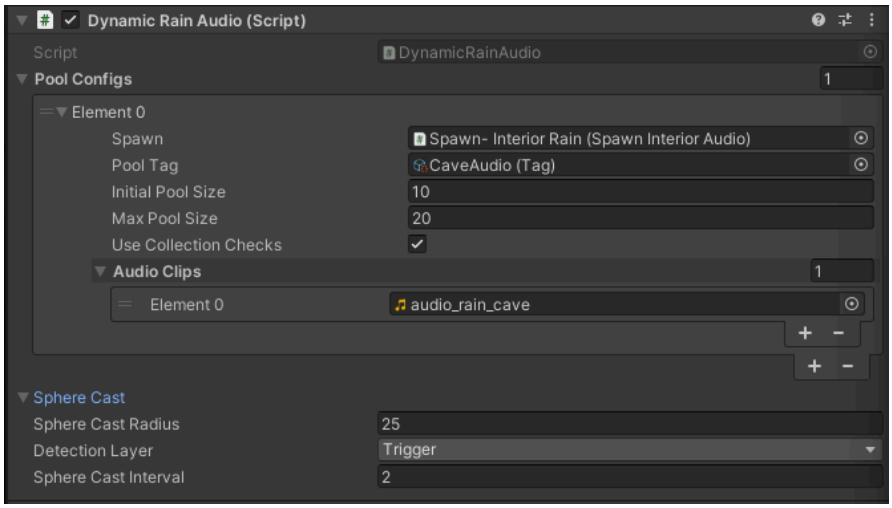
DYNAMIC: Rain Audio

A powerful system that effortlessly manages and spawns immersive interior rain audio sources within buildings and structures. This innovative tool utilizes Unity's Object Pooling and our DYNAMIC: Spawn System to optimize performance. Distributing rain audio sources to user placed spawn points, while significantly reducing the strain on your game by limiting the number of active audio sources. With the ability to create distinct audio pools for different interior types, you can easily differentiate sounds, such as cave acoustics versus interior building rain, ensuring the right ambiance for every environment. Plus, with support for multiple audio clips per pool, the system can randomly select different sounds upon spawning, helping you avoid repetitive audio and creating a more engaging experience.

Key Features

Performant Audio Management:

- Utilize Unity's Object Pooling system to distribute rain audio sources to user-designated spawn locations inside buildings and structures. This approach reduces the strain on your game's performance by limiting the number of active audio sources at any given time.



Distinct Interior Types:

- Create different audio pools for different interior types. This allows you to have for example cave audio and interior building audio, where the system will read the spawn type and spawn the correct interior rain audio.

Multi Clips Per Audio Pools:

- Lets you have multiple different audio clips per pool. This is help avoid repetitive sounds as the system will randomly select a clip when it's spawned in the scene

Spawn- Interior Rain Audio

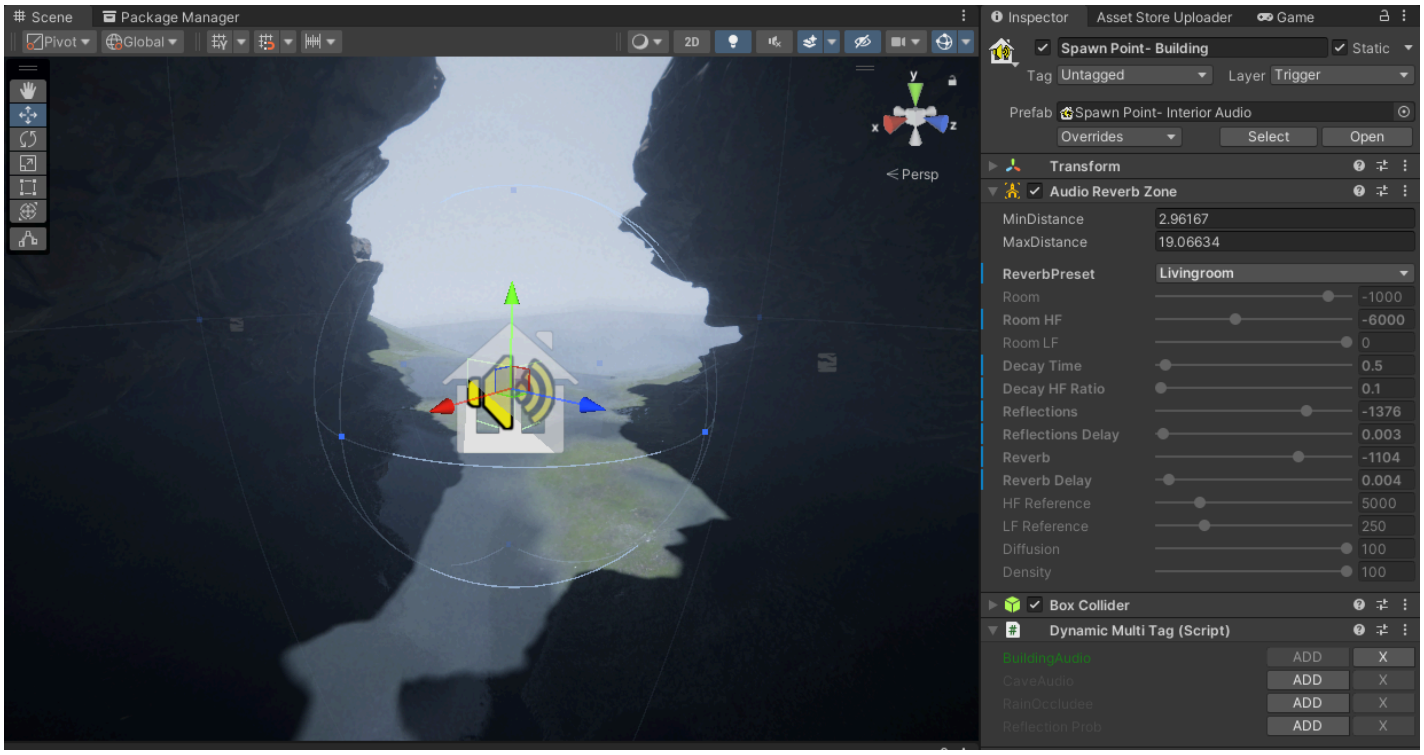
Designate specific spawn points within your game world where interior rain sounds will spawn.

Simple to Set Up:

- Simply add one of these spawn points to your gameworld and specify your interior type using the tag feature from the Dynamic- Multi Tag component and you're ready to go!

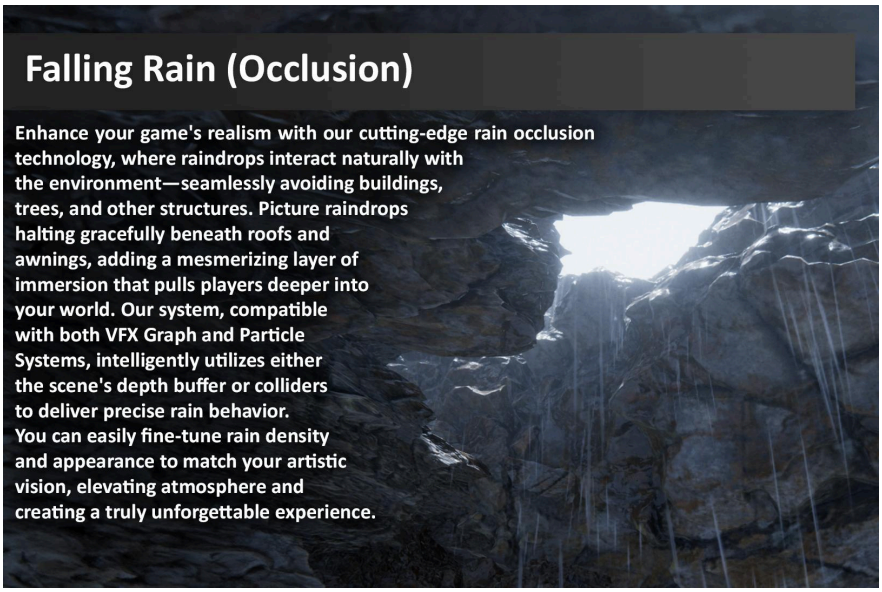
Optimized:

- Fully integrated with the Dynamic- Rain Audio so these spawn points only hold audio sources and clips when the player is near them. This heavily saves on performance and memory usage.



DYNAMIC: Falling Rain

The ultimate tool for adding unparalleled realism and immersion to your Unity projects. With full control over rain/wetness effects, this asset allows your environments to react dynamically to changing weather conditions, breathing life into your scenes like never before. Whether you're crafting a light drizzle or a torrential downpour, DYNAMIC- Falling Rain gives you complete control to create the perfect atmosphere for any scenario. From lingering puddles to wet surfaces that stay after the rain stops, this tool ensures your world feels alive and immersive.



Key Features

Customizable Look and Feel:

- Add stunning fully customizable falling rain effects to your scenes. With tons of settings which let you create the look and feel that will immerse your players.

Falling Rain Occlusion:

Achieve unparalleled realism in falling rain occlusion with our advanced occlusion systems, which are designed to work seamlessly together out of the box. Combine VFX Occlusion Zones, Camera Depth Buffer Occlusion, and SDF Occlusion Zones in any configuration—or use all three simultaneously—to create visually stunning environments. Or use our Aerial Camera Occlusion system paired with Camera Depth Buffer Occlusion if it suits your needs. For developers leveraging Unity's particle system, take advantage of its built-in occlusion feature, which interacts with your scene's colliders to occlude rain dynamically.

VFX Occlusion Zones:

- Place these boxes to block falling rain from going through gameobjects. Example... inside buildings, or under covered areas.

SDF Occlusion Zones:

- These allow you to have extremely accurate occlusion for your falling rain. These files are created by combining all your static objects in your scene into 1 big occlusion zone, which can be read by Unity's VFX Graph.

Camera Depth Buffer Occlusion:

- Uses the power of Unity and VFX Graph's built in camera occlusion, which uses your main camera depth buffer

Aerial Camera Occlusion:

- Blocks falling rain based on objects that block an aerial camera's view. This version is the legacy version and is no longer the recommend type to use

Unity's Particle System (mesh collider occlusion):

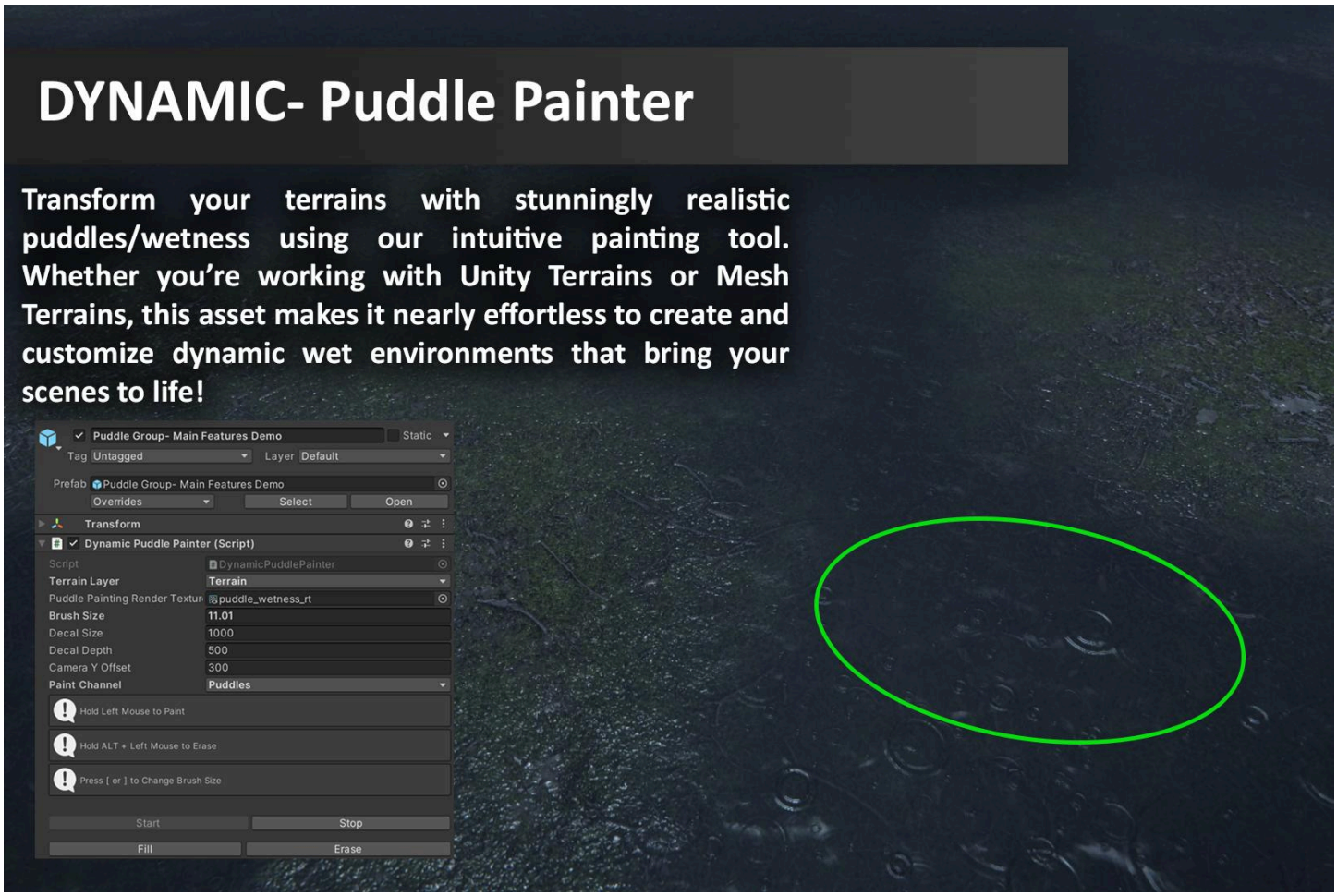
- This version uses your scenes gameobject colliders to block rain from going through gameobjects.

Performance Optimized

- Our smart system ensures the Falling Rain VFX game object always follows the player, optimizing performance in both small and large environments. The rain effect only appears within the player's visible range, reducing unnecessary processing. Combined with the built-in optimizations of Unity's VFX Graph, this system delivers a visually immersive experience without compromising performance.

DYNAMIC: Puddle Painter

Transform your terrains with stunningly realistic puddles/wetness using our intuitive painting tool. Whether you're working with Unity Terrains or Mesh Terrains, this asset makes it nearly effortless to create and customize dynamic wet environments that bring your scenes to life. And with different selectable paint layers and a customizable puddle shader you have the freedom to create the environment you desire!



Key Features

Easy Painting:

- Easily paint puddles/wetness directly onto your terrains. Draw simple shapes and watch as our powerful puddle shader transforms them into realistic puddles!

Draw Simple Shapes get Amazing Results:

- Click and hold your mouse to paint simple shapes and watch as our powerful puddle shader transforms them into realistic shaped puddles!

Multi Paint Layers:

- Choose from a range of layers to paint and mask out, allowing for precise control over your wetness and puddle effects.

Selectable Paint Layers

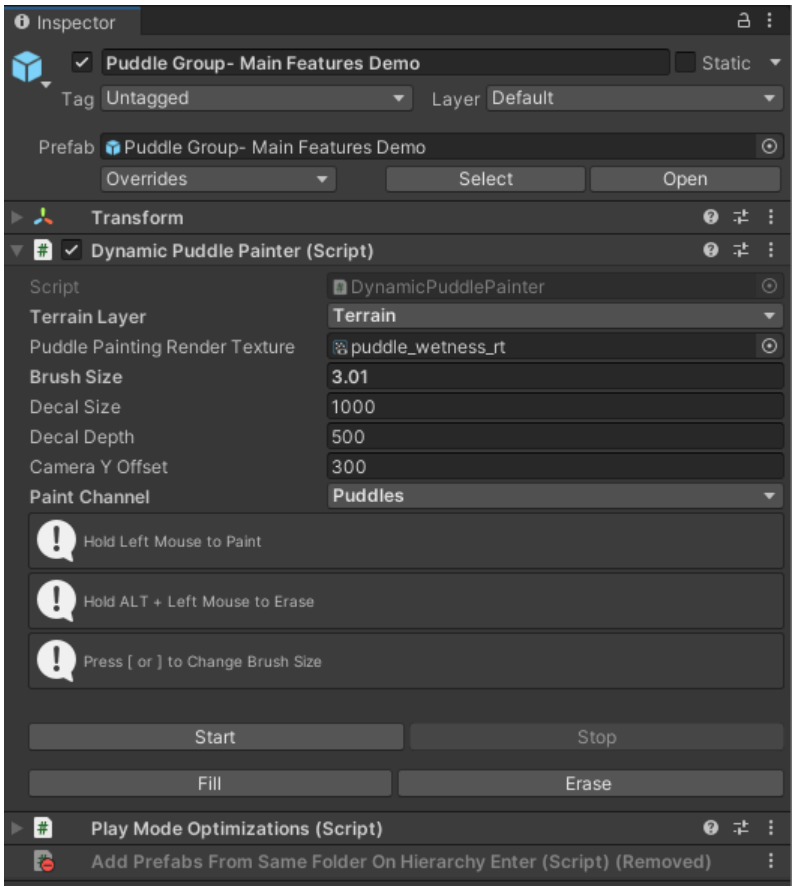
Procedural Puddles:

Power of Procedural Noise

- Cover your entire terrain with procedural puddles using noise and other shader variables. This lets you quickly cover your entire terrain in puddles without manually painting each puddle by hand.

Hex Tiling

- Uses hex tiling to help break up repetitive puddle patterns, ensuring a more organic and visually appealing look.



Wetness:

Easy Terrain Wetness

- Make your entire terrain wet with this simple shader. This feature adds a layer of realism, making surfaces appear damp and freshly rained on.

Puddles:

- Draw simple shapes to create puddles exactly where you want them. Our shader uses noise and advanced variables to turn these shapes into realistic puddles, perfectly integrated with your terrain.

Procedural Terrain Puddle Mask:

- Enhance your terrain with procedural terrain puddles by combining third-party assets like Procedural Terrain Painter and Unity's Terrain Tools to create slope and curvature masks to quickly add puddles based on the terrain's topology

All Layers

- Paint all layers at once

All Layers Minus Procedural Terrain Puddle Mask

- Paint all layers minus the Procedural Terrain Puddle Mask

DYNAMIC: Dripping Rain

Transform your game scenes into deeply immersive, rain-soaked environments with DYNAMIC - Dripping Rain—the powerful, easy-to-use system that lets you place dripping rain effects in your environments. Whether it's rain dripping from the edges of rooftops, or dripping from a character or item that's moving through the environment, this system is built to deliver an unparalleled level of immersion, realism, with total creative control and customization.

Key Features

Easy Placement:

- Easily place and customize the size of dripping rain spawn points. These can be placed on the edge of buildings, NPC's, the Player or other surfaces where rain would drip from. Allowing you to specify exactly where or what rain should drip from.

Customizable Look and Feel:

- Our system gives you the freedom to customize the look and feel of the dripping rain to your heart's content. Whether it be heavy drops falling from ledges or fine mist dripping through trees our system can create it all.

Optimized for Huge or Small Environments:

- Build the atmosphere and worlds you desire with no compromise in image fidelity and performance . As our smart spawning system and Unity Object Pooling takes care of the heavy lifting by reducing the amount of dripping rian VFX that can be in the scene at once.

Edit Mode Preview:

- Visualize the dripping rain effects directly in the editor with one-click preview, to see exactly how the effects will appear at runtime—no surprises. As the in-game spawning system only works in playmode.

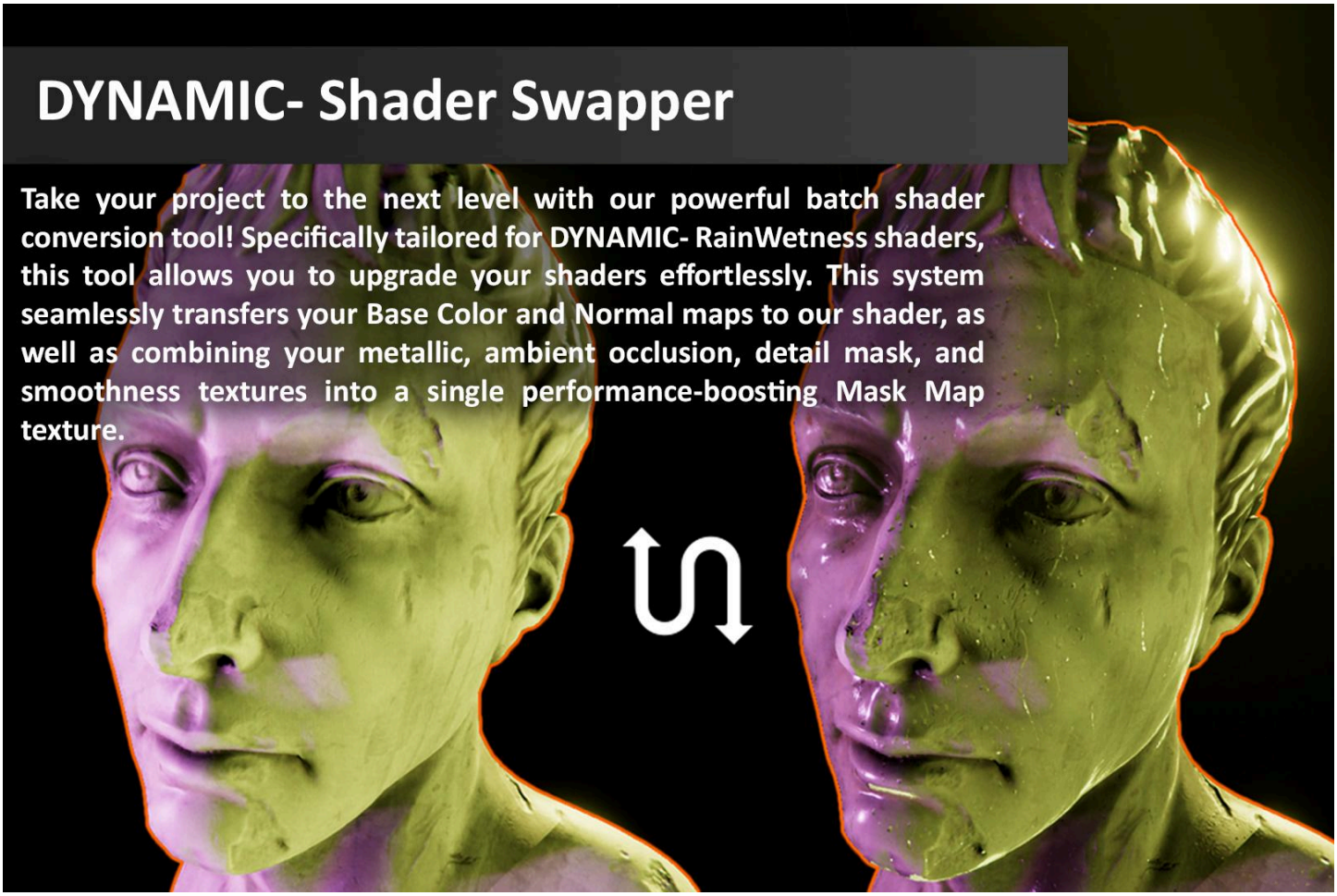
Build Mode Optimized:

- Automatically optimizes dripping rain spawn points in build mode by removing all components used for debugging.

DYNAMIC: Shader Swapper

★ Video: Swapping Shaders

Take your project to the next level with our powerful batch shader conversion tool! Specifically tailored for DYNAMIC-RainWetness shaders, this tool allows you to upgrade your shaders effortlessly. This system seamlessly transfers your Base Color and Normal maps to our shader, as well as combining your metallic, ambient occlusion, detail mask, and smoothness textures into a single performance-boosting Mask Map texture.



Key Features

Effortless Upgrade Your Shaders:

- With just a few clicks you can take your shaders to the next level and avoid the annoying hassle of manually converting each of your materials to the new shaders

Save on Draw Calls:

- Combines multiple textures into a single Mask Map texture.Significantly improves performance by reducing the number of texture samples and draw calls required.

Batch Conversion:

- Saves you time by letting you add endless shaders for conversion.
    - Example: setup conversions for shaders A⇒ B, B⇒C, and A⇒C

Fix Shader Errors:

- This tool can also help you fix the “Hidden Shader Error”. Which can happen when converting your assets from URP to HDRP

DYNAMIC: Shader Property Finder

This powerful tool simplifies the process of identifying texture variables in your shaders, making your life easier WHEN swapping shaders using the Dynamic- Shader Swapper tool.

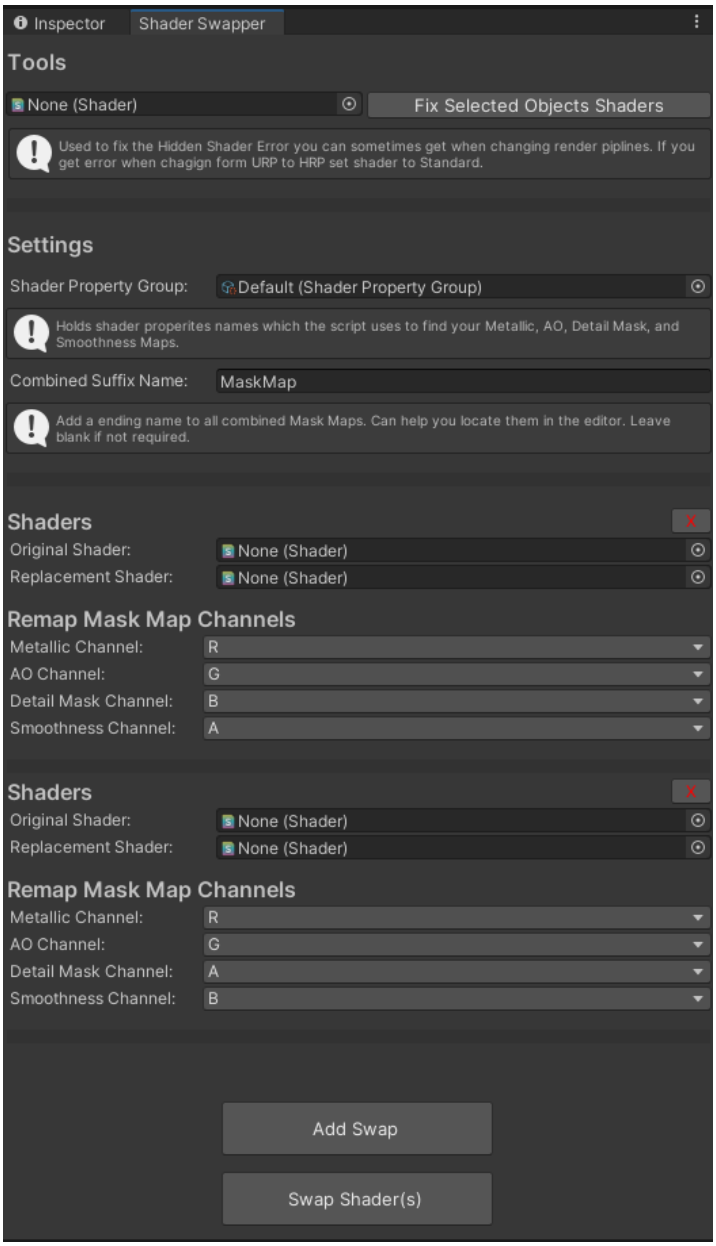
Key Features

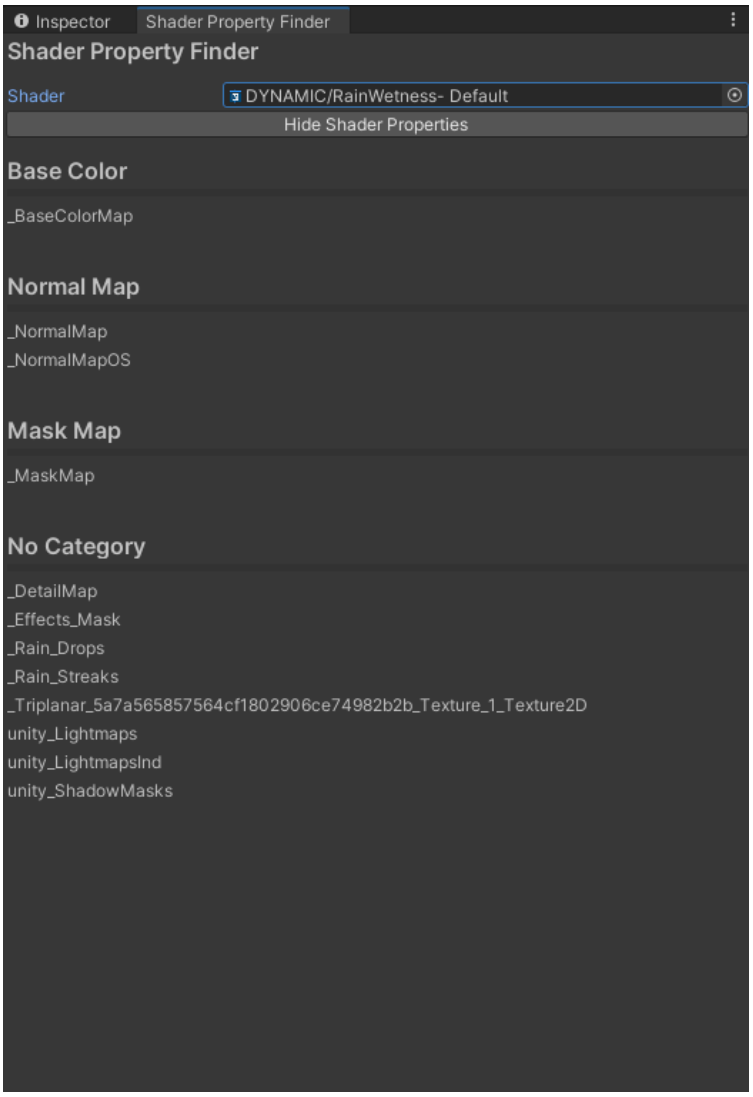
Automatic Texture Variable Finder:

- Effortlessly scan your shader for texture variables which can then be added to your Shader Property scriptableobject

Color Coded:

- As of version 1.1 the system has a variable to specify your Shader Property scriptable object. The updated system will cross reference that object and your shader and let you know what variables it has or is missing via green and red color respectively

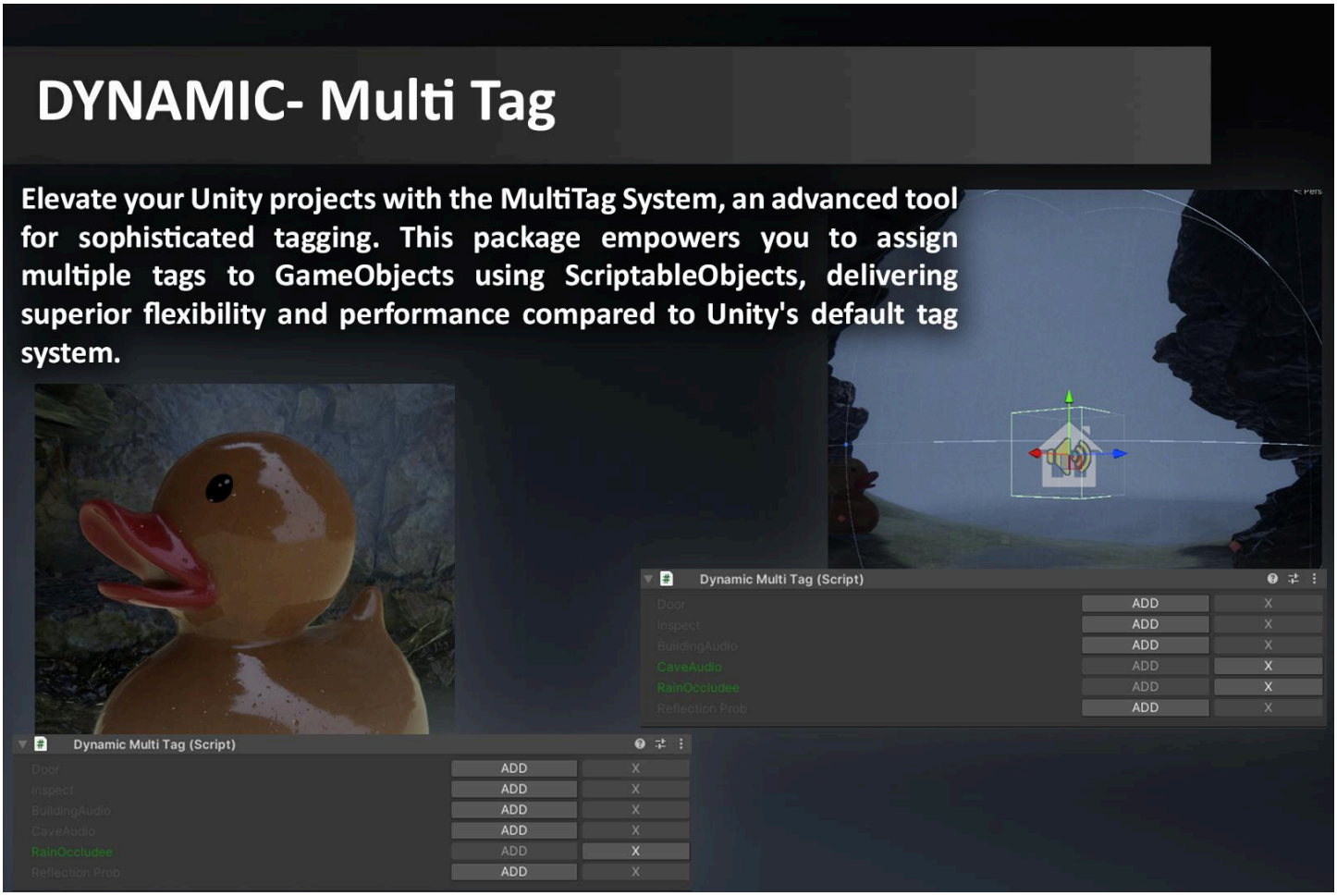




DYNAMIC: Multi Tag

★ This is a Third Party asset that has been altered for use with DYNAMIC- RainWetness

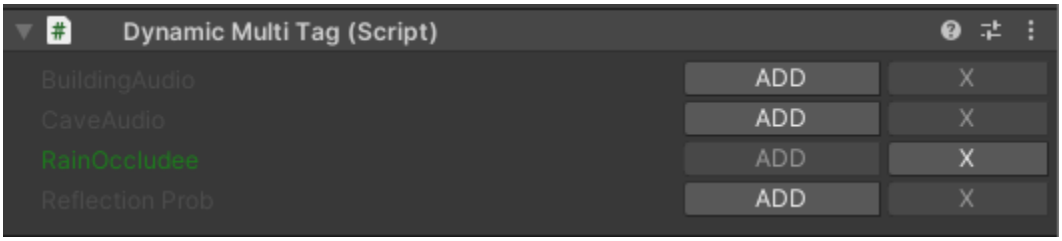
An advanced tool for sophisticated tagging. This system empowers you to assign multiple tags to Gameobjects using ScriptableObjects. Delivering superior flexibility and performance compared to the default tag system.



Key Features

Flexible Multi or Single Tagging:

- Easily assign single or multiple tags to GameObjects. The system supports dynamic tag addition and removal through both code and a custom editor GUI, offering streamlined tag management that adapts to your project's needs.



Unmatched Performance:

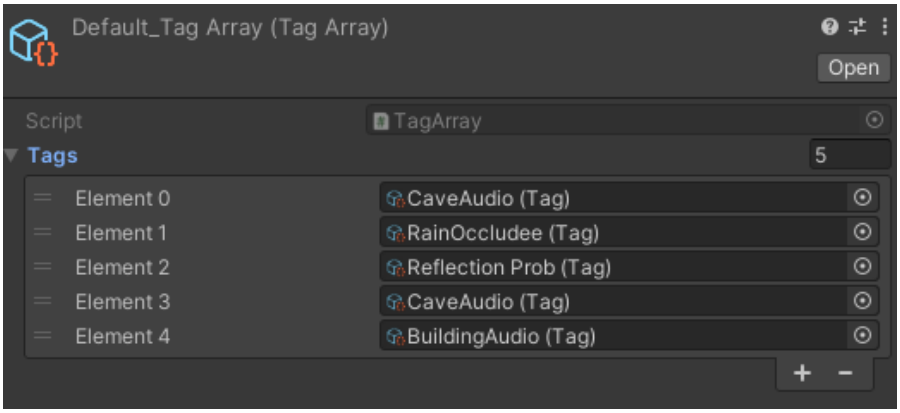
- Benchmarks show that this system performs tag checks in approximately 1 millisecond, which is up to 11% faster than Unity's default tag system, which typically takes around 11 milliseconds. This exceptional performance ensures smooth and responsive gameplay, even in complex scenes with numerous GameObjects.

Custom Editor GUI:

- Leverage a custom editor GUI for effortless tag management. This intuitive interface allows you to add and remove tags for each GameObject directly within the Unity Editor, simplifying the process and eliminating the need for manual code adjustments.

Tag Arrays:

- Organize and manage different sets of tags with Tag Arrays. This feature lets you combine multiple tags into a single asset, effectively categorizing and utilizing distinct groups of tags for various project needs.
- **Example of Tag Arrays:**
    - Enemies Tag Array: Includes tags such as "EnemyType1," "EnemyType2," and "Boss."
    - Allies Tag Array: Includes tags like "FriendlyNPC," "PlayerCharacter," and "NeutralAlly."



Improved Over Find Game Objects With Tag:

- The MultiTag System outperforms Unity's FindGameObjectsWithTag method, which is limited to single tag searches and can be inefficient with numerous GameObjects. This package supports multiple tags and tag arrays, enabling more precise queries and better performance. By using Tag Arrays and the runtime API methods, you can manage and interact with GameObjects more effectively, reducing overhead and enhancing overall game performance.

DYNAMIC: Systems Manager

★ **Only one instance of this script can exist per scene. This ensures that all variables are shared across all scripts.**

This script is designed to manage and streamline essential global components in your project. It automatically detects and assigns key components such as the Player GameObject, main AudioSource, and dynamic weather and occlusion systems, ensuring they are always accessible and correctly initialized.

Key Features

Time-Saving:

- Centralizes the management of global components, reducing repetitive code to increase overall performance.

Do Not Destroy On Load:

- Keep all DYNAMIC: Rain/Wetness scripts and system alive when switching scenes.

Enable/Disable All Debug Gizmos:

- Easily enable/disable all debug gizmos for all DYNAMIC scripts, saving you time.

DYNAMIC: Wind (with Global/Local Fog Integration)

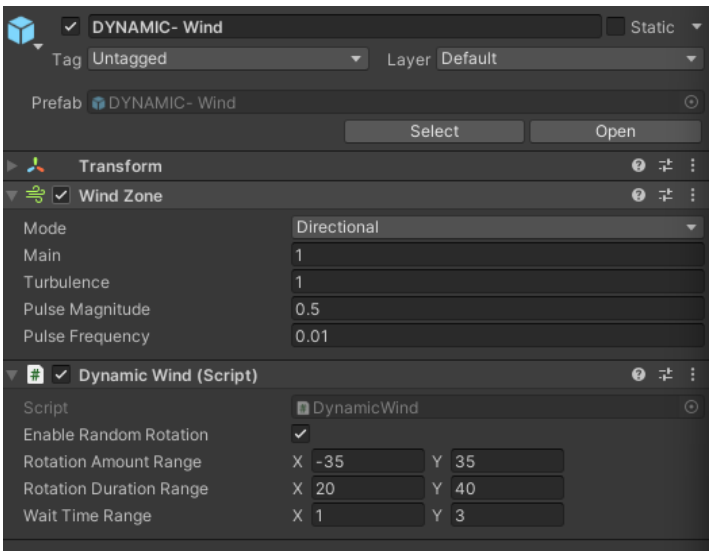
Watch as fog and falling rain dynamically respond to changes in wind direction, creating a truly immersive experience. Perfect for any weather system, our DYNAMIC- Wind system ensures your world feels alive and responsive. Immerse your players in a dynamic, ever-changing atmosphere!



Key Features

Global Rain Value Integration:

- The fog's density and behavior adjust in real-time based on the global rain value. As rain intensifies, fog becomes denser, enhancing the atmosphere, while it gradually dissipates as the rain subsides, creating a realistic and immersive weather experience.

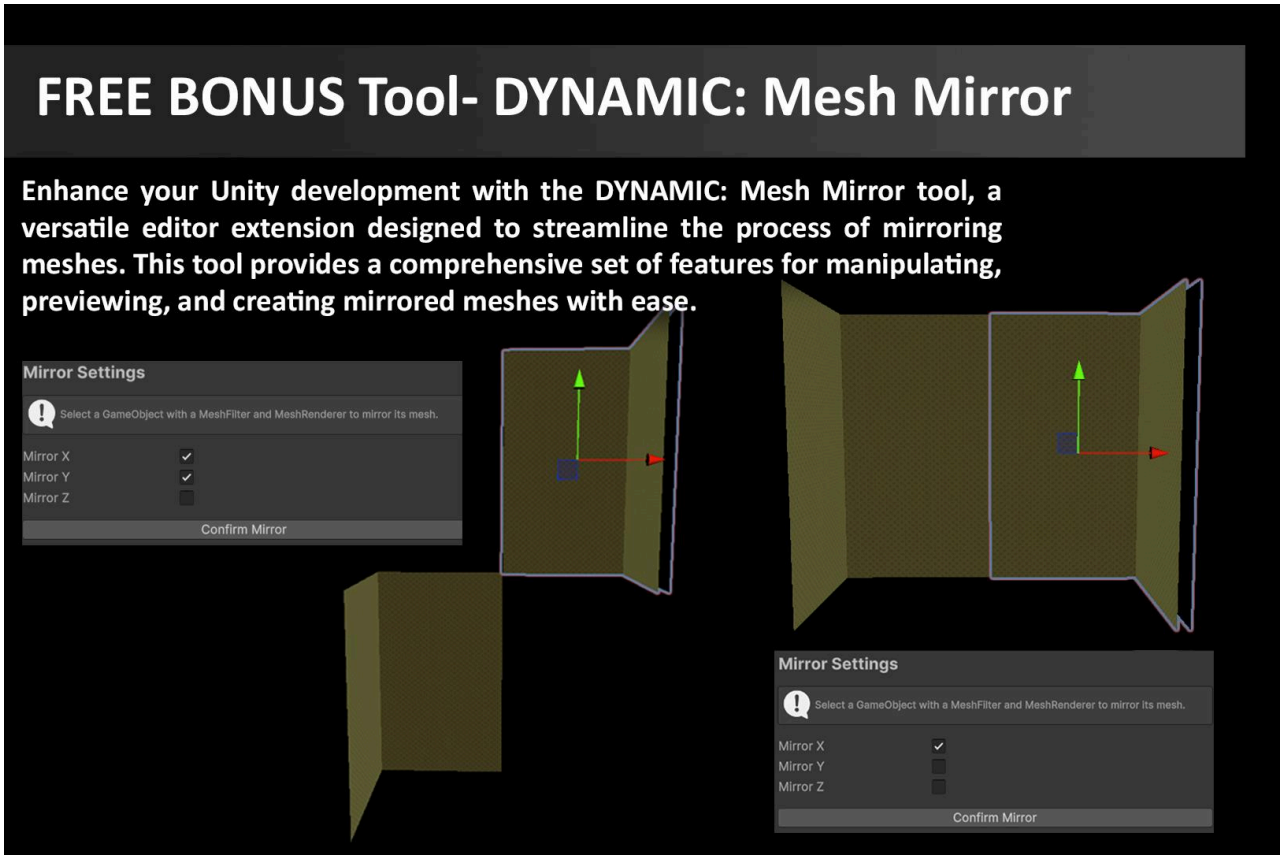


Fog Response to the Wind Direction :

- Integrated with dynamic wind systems, the volumetric fog shifts and moves in sync with the wind direction. This interaction adds a layer of realism, with fog visibly drifting and swirling as the wind changes, perfectly complementing the environmental conditions of your scene.

DYNAMIC: Mesh Mirror

Enhance your Unity development with the DYNAMIC: Mesh Mirror tool, a versatile editor extension designed to streamline the process of mirroring meshes. This tool provides a comprehensive set of features for manipulating and previewing mirrored meshes with ease.



Key Features

Mirror Axis Selection:

- Choose from various mirror axes, including X, Y, Z, or combinations thereof (XY, XZ, YZ, XYZ), to precisely control how your mesh is mirrored.

Transformation Scaling:

- Optionally apply scaling transformations to the mirrored mesh. Adjust the transformationScale to modify the size of the mirrored output, offering flexibility in how the mirrored mesh fits into your scene.

UV Mirroring:

- Toggle UV mirroring to ensure that texture coordinates are adjusted accordingly during the mirroring process. This ensures that your textures align correctly with the mirrored geometry.

Preview Functionality:

- Visualize changes before finalizing them. The tool provides a live preview of the mirrored mesh, allowing you to make adjustments and see the results in real time.

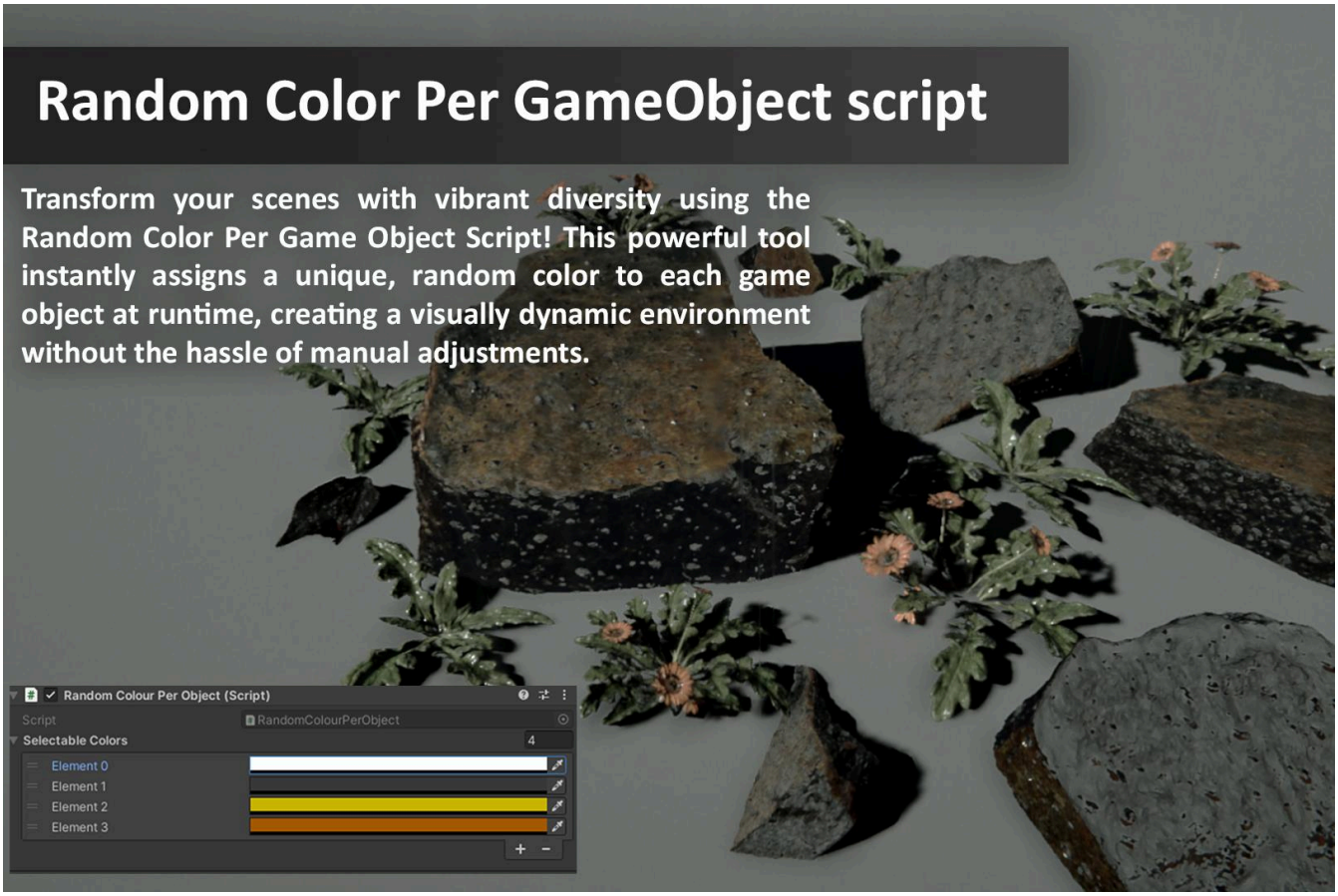
Save Options:

- Choose to save the mirrored mesh either as a new GameObject and mesh or overwrite the existing mesh. This feature is useful for keeping your original assets intact while experimenting with different mirrored versions.

Random Colour Per GameObject

★ **Works with all DYNAMIC- RainWetness Shaders**

The ultimate tool for dynamic, runtime color variation. This automatically assigns a random color from a user defined color palette to each game object in your scene at runtime. This helps to easily add uniqueness and help avoid repetitive looking assets like rocks, or trees. This feature is already included in all Rain/Wetness Shaders; you just have to add 1 script to your gameobject and you're ready for effortless randomization.



Key Features

Dynamic Color Assignment at Runtime:

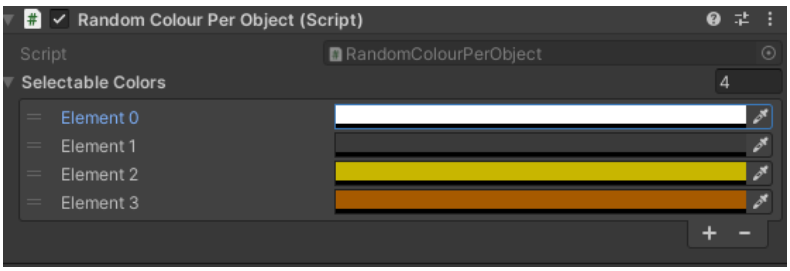
- Automatically assigns a random color to each game object from a predefined array of colors at runtime, ensuring no two objects look the same.

Automatic Execution:

- The script performs its function of assigning colors at runtime and then destroys itself, keeping your scene clean and optimized.

Color Variety Per Gameobject:

- Perfect for scenarios such as assigning random color variations to rocks, trees, or any repetitive objects in your scene, adding a touch of uniqueness and realism.



Updating Package

1. Prepare for Update

1. Before updating to the new version please make sure to make a copy of your ENTIRE MoveRotateResize folder and paste it to your desktop or another location
2. Once that is completed please delete the entire MoveRotateResize folder that's inside of your Unity project
3. Now you can update and install new version of the asset

2. Import Package

1. Import the DYNAMIC: RainWetness package from the package manager
2. If a "Script Updating Consent" box pops up press "No"
3. Now you're ready to go!

Updating to v1.3.2

★ **!!STOP!! Process for updating to new version**

1. Before updating to the new version please make sure to make a copy of your ENTIRE MoveRotateResize folder and paste it to your desktop or another location
2. Once that is completed please delete the entire MoveRotateResize folder that's inside of your Unity project
3. Now you can update and install new version of the asset

Updating to v1.3.1

★ **!!STOP!! Process for updating to new version (if updating from version 1.3)**

1. Simply download and install hotfix. You DON'T need to delete your folder this time.
2. Go to Tool/MoveRotateResize/ DYNAMIC- RainWetness/First Time Setup/Setup. Press skip on the shader install.

★ **!!STOP!! Process for updating to new version (if updating from version 1.2 or lower)**

1. Before updating to the new version please make sure to make a copy of your ENTIRE MoveRotateResize folder and paste it to your desktop or another location
2. Once that is completed please delete the entire MoveRotateResize folder that's inside of your Unity project
3. Now you can update and install new version of the asset

Updating to v1.3

★ **!!STOP!! Process for updating to new version**

- Version 1.3 is a big update and changes/optimizes many systems. As a result you'll have to delete your old DYNAMIC- Rain Wetness Systems gameobject in your scenes and add a new one. Sorry for the inconvenience.

★ **!!STOP!! If you used the puddle painter**

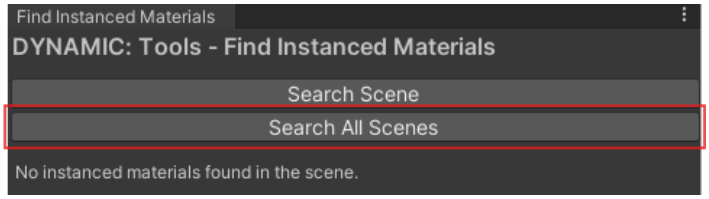
- The "rt_puddle_and_wetness" render texture was increased from 1024 x 1024 to 2048 x 2048 resolution. As a result you will have to either delete your Puddle Paint Mask texture for each of your custom painted terrain or change the "rt_puddle_and_wetness" render texture back to 1024 x 1024

Updating from v1.1 or v1.1a

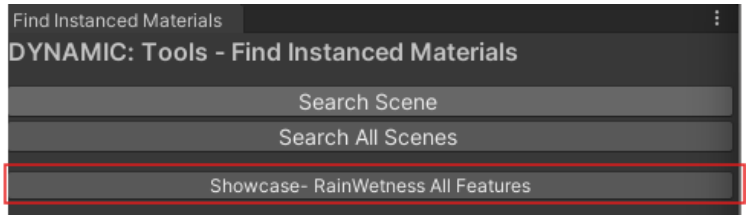
★ **If you used "Edit Mode Occlusion" from version 1.1 or 1.1.a**

An error in the previous version of Edit Mode Occlusion was causing gameobjects materials to permanently become instances, making the materials uneditable. The steps below fix this issue, and the new refactored version of the script prevents that from happening in the future. Very sorry for the inconvenience.

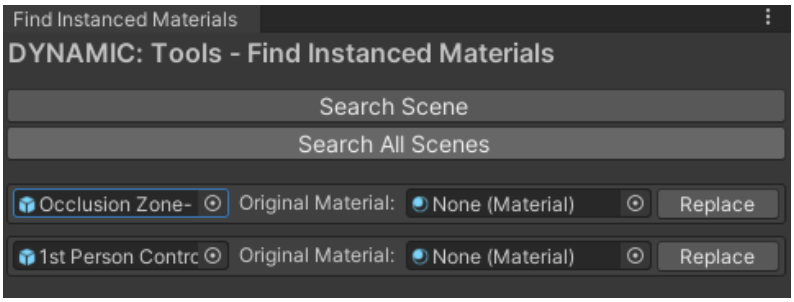
1. Go to each of your scenes that you used “Edit Mode Occlusion” and do the following...
2. Go to Tools/MoveRotateResize/DYNAMIC: Tools/Find Instanced Materials



3. Press Search All Scenes
4. A list of buttons for each scene that’s affected will appear
5. Click the button for each scene to load it



6. Once a scene is loaded press “Search Scene” to check for instanced materials
7. A list of materials that are instanced will appear
    1. The system tries to auto find the original material, if it can’t it’ll say “None” in the material slot
    2. If it says “None” then click on the gameobject to jump to it
    3. Then replace the material manually with the correct one from your project folder
8. If the system auto found an original material and it looks correct then press “Replace”
    1. Rinse and repeat this process thill no more scenes with the affected instanced materials can be found



Installing Package

1. Remove Old Folder

★ !!STOP!!

1. Before updating to the new version please make sure to make a copy of your ENTIRE MoveRotateResize folder and paste it to your desktop or another location
2. Once that is completed please delete the entire MoveRotateResize folder that's inside of your Unity project
3. Now you can update and install new version of the asset

2. Import Package

1. Import the DYNAMIC: RainWetness package from the package manager
2. If a "Script Updating Consent" box pops up press No

3. First Time Setup

1. Go to... Tools/MoveRotateResize/DYNAMIC- RainWetness/First Time Setup/Set Up
2. Click Install Shaders or Skip
    - This auto installs the correct RainWetness shaders for your Unity version
    - Skipping is only recommended if you don't want the shaders to be changed. Example you customized them in a previous version
3. If a "Script Updating Consent" box pops up press No
4. The setup script will now auto update scripting definitions, and downloadunity's TextMesh Pro, VFX Graph, Shader Graph, Unity's Surface Gradient Shader 2.1, Naughty Attributes, and Poly Brush

4. Enable Input Systems

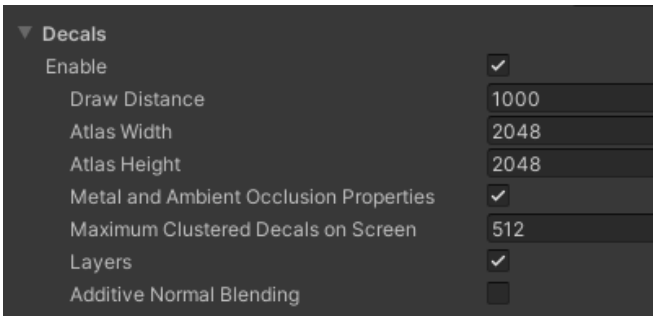
1. Go to Edit/Project Settings
2. Then click on Player and under Configuration you'll see Active Input Handling*
3. Change this to Both or just use the old input system (as this asset currently requires the old system)

5. Enable Decals

★ **DON'T enable Adaptive Normal Blending as this will make the puddles look incorrect**

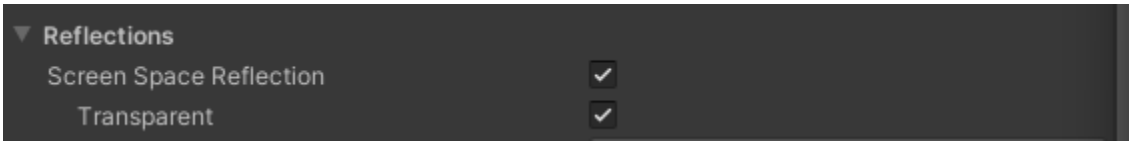
★ **Puddles use Decal Layer 1**

1. Go to... Edit/Project Settings/Quality/HDRP
2. Enable Decals and Decal Layers

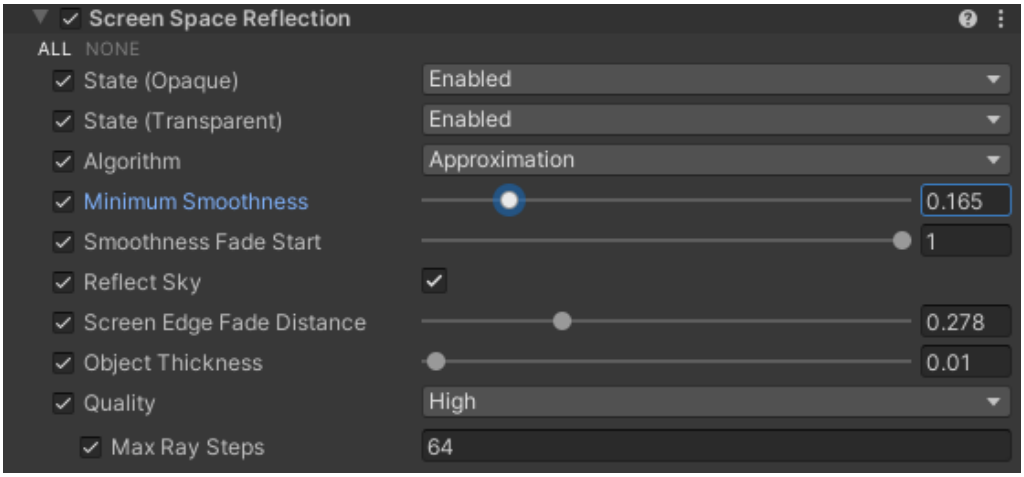


6. Enable Reflections

1. Go to... Edit/Project Settings/Quality/HDRP
2. Under Reflections Enable
3. Screen Space Reflections
4. Transparent

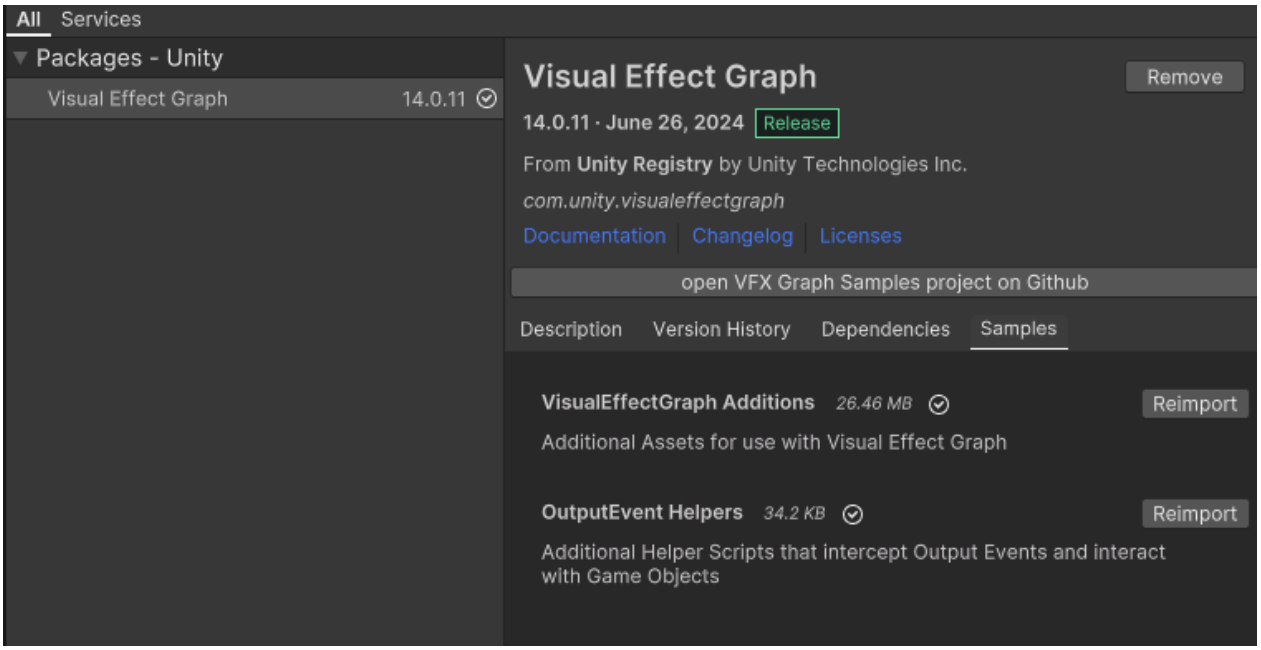


5. In all of your Post Processing volume under Screen Space Reflections set the Minimum Smoothness slider to a low value



7. Import Visual Effects Graph Additions

1. Go to... Window/Package Manager
2. Search Unity Registry for Visual Effects Graph
3. Click on the Samples tab
4. Click import VisualEffectsGraph Additions and Output Event Helpers



8. Add Layers

★ **These are required to make all the demo scenes work correctly**

1. Go Layers/Add Layer...
2. Then name the following layers...
    1. 29 =Characters
    2. 30 = Trigger
    3. 31 = Painter

Instructions

Converting Shaders

Shader Swapper

★ ***Video:*** ***Swapping Shaders***

Location

- Go to...Tools/MoveRotateResize/DYNAMIC- Shader Swapper/Shader Swapper

Tools Section

Fixes the Hidden Shader Error which can sometimes happen when converting assets from one Render Pipeline to another.

1. Assign a shader
2. Select the gameobject(s) you want to change
3. Click “Fix Selected Objects Shaders”

Settings Section

The Shader Property Field is a scriptable object that lets the Shader Swapper system know what texture properties to look for when trying to convert your shader.

1. Assign a Shader Property Group scriptable object
    a. You can make a new one by copying an old one or create a new one by going to Tools/MoveRotateResize/DYNAMIC- Shader Swapper/Shader Properties
2. Type a name into the suffix field
    a. This will be added to the end of each combined Mask Map
    b. This isn’t required but can be handy to help you identify combined textures

Shaders Section

Use this section to convert your shaders to any DYNAMIC Shader. This system auto assigns your Base Color and Normal maps to the new shader as well as combines your Metallic, AO, Detail Mask, and Smoothness maps into 1 Mask Map Texture.

1. Add your shader to the original shader field
2. Add one of the DYNAMIC Shaders to the replacement field
    a. If applicable Click Add Swap to add another shader you’d like to convert

i. This system allows you to convert multiple different shaders at the same time
    b. If applicable Click the red X to remove a swap

3. Click Swap Shader(s)
    a. This process might take a while depending on the amount of materials you’re converting

★ **You’ll have to manually adjust all your shader(s) variables, as this system only swaps textures**

★ **If you get a Ran Out of Memory Error Just press ok and restart Unity. As this happens if your system is low on harddrive space**

Shader Property Finder

★ **DO NOT delete any of the default variables, as these are required for the script to work correctly**

Location

- Go to... Tools/MoveRotateResize/DYNAMIC- Shader Swapper/Shader Property Finder

Steps

If your shaders aren't converting properly when using the DYNAMIC Shader Swapper use this to help identify your shaders texture variables.

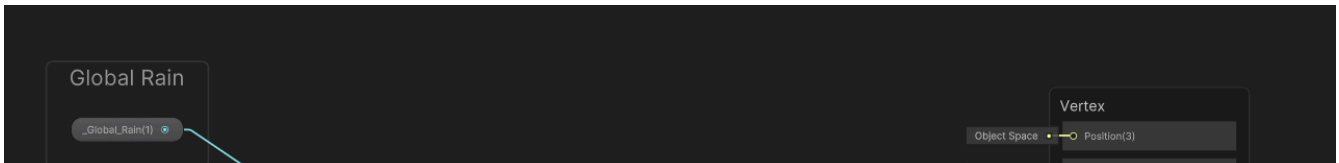
1. Add the shader you want to inspect to the field
2. The window will then populated with all of the shaders texture variables and split them into categories

Adding Rain/Wetness to your own

Shader Graphs

1. Go to folder...Assets\MoveRotateResize\DYNAMIC- RainWetness\Shaders\Sub Graphs\RainWetness\RainWetness Systems
    - Open RainWetness Systems (Copy Me)
    - Drag select all variables and the sub-graph and press copy
2. Go to your Shader Graph shader
3. Paste the variables and sub-graph into a blank space
4. Drag select all copied variables then right click then go to Convert to... then press Property
5. Rename all copied variables to match variable names from the RainWetness Systems (Copy Me) shadergraph
6. Make sure to uncheck exposed on some variables so that they match the ones from the RainWetness Systems (Copy Me) shadergraph
7. Connect all links to their appropriate inputs and outputs

Not Exposed	**Exposed**
• _Global_Rain • _Global_Rain_Drop_Size • _Global_Rain_Drop_Size • _Global_Rain_Drop_Speed • _Global_Rain_Streak_Size • _Global_Rain_Streak_Speed • _Global_Rain_Streak_Distortion_Strength	• Effects Mask • AWSM- Top/Bottom- Distance • AWSM- Top/Bottom- Softness • Rain_Occlusion • Non-Porus/Porus • Object Smoothness Multiplier • Object Normal Smoother • Rain Drops • Dynamic Object • Drop Size (Dynamic Object) • Drop Normal Strength • Drop Smoothness Strength • Rain Streaks • Streak Normal Strength • Normal Distortion Strength • Streak Smoothness Strength

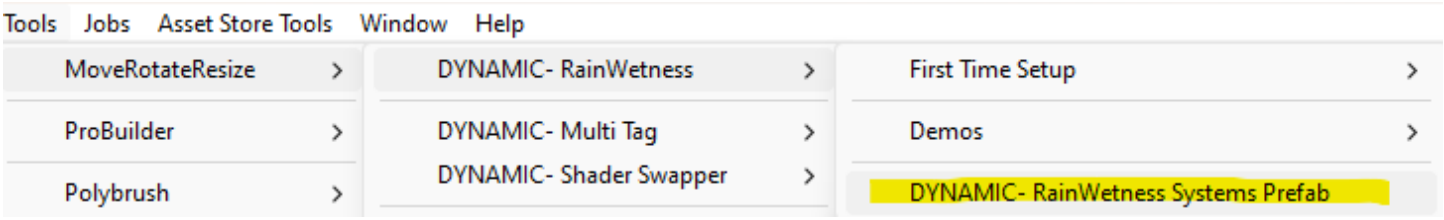


DYNAMIC: RainWetness Systems prefab

This is the main home for all of your Rain/Wetness systems allowing you to control most settings from this one location. Below is a breakdown of each script/manager that this gameobject holds.

Adding to your Scene:

1. Go to MoveRotateResize/DYNAMIC- RainWetness and click on DYNAMIC- RainWetness Systems Prefab
2. Finished! You now have all rain/wetness systems added to your scene



DYNAMIC: Systems Manager

Do Not Destroy On Load:
- Enable this to have DYNAMIC- RainWetness Systems beagle to move between scenes

Set Debugs All True:
- This will turn all debug visuals on for all DYNAMIC scripts

Set Debugs All False:
- This will turn all debug visuals off for all DYNAMIC scripts

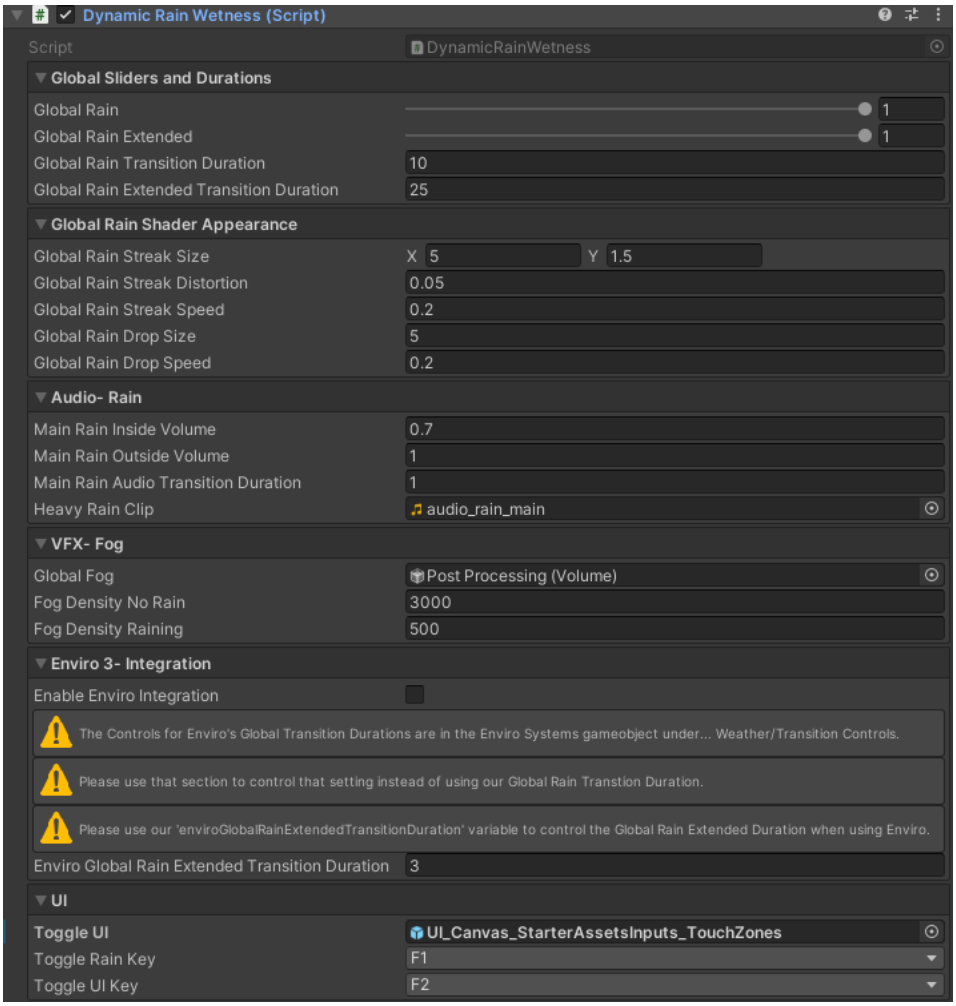
DYNAMIC: Rain Wetness

Global Sliders and Durations:

3. The Global Rain and Global Rain Extended sliders control the global coverage for all shaders.
    - These variables can be controlled via code in playmode
    - They can also be dragged in editor to update in edit mode
4. Set the transition duration of Global Rain and Global Rain Extended variables to control how long each one takes to transition in seconds

Global Rain Shader Appearance:

- Controls global settings for all Rain/Wetness shaders...
    - **Global Rain Streak Size**
    - **Global Rain Streak Distortion**
    - **Global Rain Streak speed**
    - **Global Rain Drop Size**
    - **Global Rain Drop Speed**



Audio- Rain:

5. **Main Rain Inside Volume**
    - This will be the volume that it changes to when the player is inside an Occlusion Zone
6. **Main Rain Outside Volume**
    - This will be the volume that it changes to when the player is outside of an Occlusion Zone
7. **Main Rain Audio Transition Duration**
    - This is the duration in seconds that it takes for the audio level to change when the player goes inside/outside of an Occlusion Zone
8. **Heavy Rain Audio Clip**
    - This is the main rain audio clip that will play when it's raining

VFX- Fog:

9. **Global Fog**
    - Place your Global Post Processing Volume here that you want controlled by our systems
    - This currently only affects the Fogs Density
10. **Fog Density No Rain**
    - Controls the fog density in the Global Fog post processing volume when there's no rain
    - This value will transition between Fog Density No Rain and Raining, based on Global Rain value
11. **Fog Density Raining**
    - Controls the fog density in the Global Fog post processing volume when it's raining
    - This value will smoothly transition between Fog Density No Rain and Raining, based on Global Rain value

Enviro 3- Integration:

12. **Enable Enviro 3 Integration**
    - Enable this to have our systems work with Enviro's Weather Presets.
13. **Enviro Global Rain Extended Transition Duration**
    - If using Enviro please use this value instead of the regular Global Rain Extended Transition Duration

★ **The regular Global Rain/Global Rain Extended Transition Durations under the "Global Sliders and Durations" section won't work when Enviro Integration is enabled. Instead Global Rain Transition Duration is controlled from the Enviro Systems Manager under Weather/Transition Controls and the extended value is controlled from here**

UI:

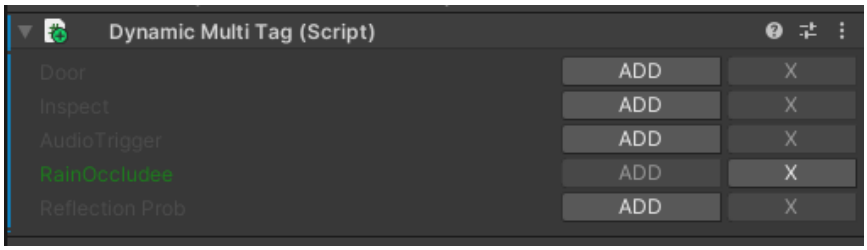
14. **Toggle UI**
    - This is the UI that will be toggled on/off with the Toggle UI Key
15. **Toggle Rain Key**
    - This will make the rain start transitioning from its current value to the opposite value (aka no rain to all rain/wetness effects turned on) . For Example it simulates your gameworld going from no rain/wetness to be completely covered with rain/wetness over time
16. **Toggle UI Key**
    - This will hide the demos UI that describes what each demo is showcasing

DYNAMIC: Occlusion and Occlusion Zones

Add Rain Occlusion to Moveable Gameobjects

★ **Make sure that your gameobject is using one of our Rain/Wetness shaders, or you have setup the Rain/Wetness Systems sub-graph on your own shader graph shader that your gameobject is using**

1. Select your gameobject
2. Add the Dynamic Multi Tag script
3. Click ADD beside the RainOccludee tag
4. Make sure your gameobject has a collider of any type and a rigidbody
5. Finished! Your gameobject will now be dynamically covered in rain/wetness when they move in/out of Occlusion Zones



Placing Occlusion Zones

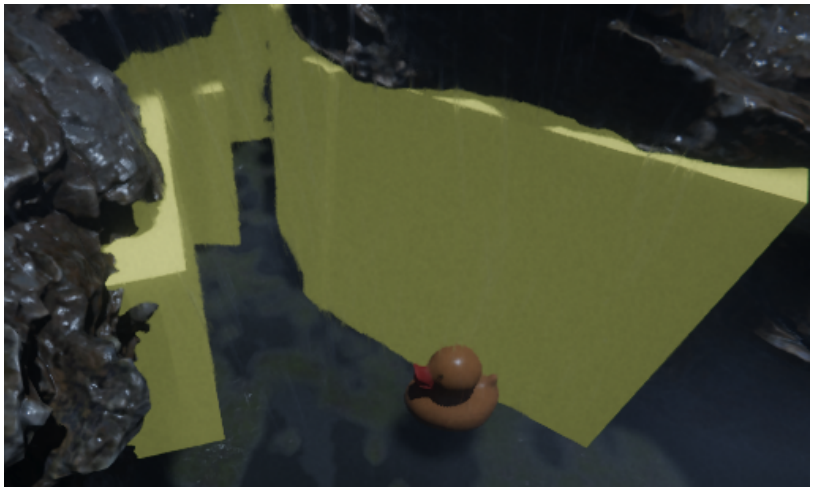
★ **As of version 1.3 DON'T rotate Occlusion Zones (please leave rotation at 0,0,0). As the new VFX Occlusion Zone system doesn't support collider rotation at the moment**

1. Select your DYNAMIC: RainWetness Systems gameobject in your scene
2. Expand it to find the Occlusion Zones gameobject
3. Duplicate the Occlusion Zones gameobject and place them in areas of your scene that you want rain to be occluded (sheltered) from rain
4. That's it. Now any gameobject that has been setup as a Rain Occludee will dynamically be covered in rain/wetness when moving in and out of these zones

Occlusion Zone Material

★ **The material for occlusion zones is only visible in the editor, as long as the zones are set to the Trigger layer and that layer isn't visible by the main camera.**

- Control the visibility of all Occlusion Zones by using the slider in the DYNAMIC: Occlusion script.
- You can set depth test on the material to LessEqual for normal viewing or set to NotEqual to make Occlusion Zones visible through geometry



Enable Edit Mode Occlusion

Pressing this button allows your objects to be affected by rain occlusion in edit mode.

★ **In Edit Mode textures of the Occluded object(s) can't be edited. Please Turn off Edit Mode Occlusion to update gameobjects textures**

★ **Edit Mode Occlusion doesn't support transitions for applying rain, it does it instantly**

★ **Edit Mode Occlusion currently only supports Shader Occlusion**

1. Make sure Editor Gizmos are enabled
2. Enter then exit Playmode (this only has to be done once every time you open Unity)
3. Press "Enter Edit Mode Occlusion" to activate
4. Press "Exit Edit Mode Occlusion" to deactivate

SDF Occlusion Zone

This type of Occlusion is used by the DYNAMIC: Falling Rain system to prevent rain from going through gameobjects. This is useful for preventing rain from going through complex shapes like the rocks that make up the cave in the Rain Audio demo scene.

★ **Currently only 1 SDF Occlusion Zone can exist in your scene at a time. Please do the following sets to combine multiple game objects into 1 SDF Occlusion Zone**

1. Make sure Editor Gizmos are enabled

DYNAMIC: Rain Audio

Creating New Spawn Pool:

1. Click the Create New Spawn button in the inspector

Adding New Trigger Tags:

★ **Please refer to the DYNAMIC- Multi Tags section for more info how tags and tag arrays work**

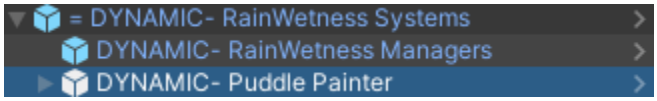
1. Go to... Tools/MoveRotateResize/DYNAMIC- Multi Tag/Tag
2. This will create a new tag
3. Rename to desired name
4. Search for Default_TagArray in your project folder
5. Select it and press + to add a new element
6. Add your tag to that new element

Placing Spawn Points:

1. Go to... Tools/MoveRotateResize/DYNAMIC- Pooling/Spawn Points/Interior Audio
2. Select your Spawn Point
3. Click ADD beside your desired tag
4. Adjust reverb zone to your desired settings
5. Move the Spawn Point to your desired location

DYNAMIC: Puddle Painter

Location:



Setup Instructions:

1. Select all of your Terrains
    a. Add a Terrain Layer
    b. Assign all of your terrains to that layer
    c. Assign Decal Layer 1 to the Render Layer Mask of your Terrain Settings
    d. Make sure your Terrain's material receives Decals
2. Search and Select Camera gameobject
    a. Add a layer named Paint Layer
    b. Assign the Layer to the culling mask
3. Remove all other layers
4. Select your scenes Main Camera
    a. Assign the Terrain Layer
    b. Unassign the Paint Layer from your Main Camera
5. Create a new Puddle Group

a. Go to...MoveRotateResize/DYNAMIC- Puddle Painter/Create Puddle Group
b. Click on the new spawned gameobject in your Hierarchy
c. Assign Terrain Layer
d. Assign Decal Size to cover your entire Terrain
e. Assign Decal Depth to encompass entre Terrain Depth and Height
f. If puddles and wetness aren't visible on your terrain you might have to change the Rendering Layer Mask of each of them to include Decal Layer 1

Painting Instructions:

Search and Select DYNAMIC- Puddle Painter game object and look for the Dynamic Puddle Painter script

1. Change your Paint Channel to the desired channel
2. Press the Start button to being painting or erasing
3. Press the Stop button to stop paint or erasing
4. Hold Left Mouse to Paint
5. Hold ALT + Left Mouse to Erase
6. Press [or] to change the brush size
7. Press Fill or Erase to fill the whole selected channel with black or white
    a. As the system uses black and white for masking layers

★ **To understand what each paint channel does click here**

Terrain Puddles/TerRainWetness Material Instructions:

★ **DON'T change either materials Draw Order**

1. Search and Select either Puddles or Wetness
2. Adjust the sliders and values to achieve the look and feel that you desire
3. Titling in the puddles material is only used for the Procedural Puddles, all other settings affect the overall puddle look
4. The Puddle Procedural Mask is used to mask out puddles when you use a 3rd party asset to create say a terrain slope mask
5. The Puddle Paint Mask is used when painting on the terrain

DYNAMIC: Falling Rain

Choose your Falling Rain Occlusion Type:

1. Use the dropdown to pick what type of Falling Rain/Occlusion type(s) you want to use...

    - **Occlusion Zones**
    - **SDF Occlusion Zone**
    - **Depth Buffer**
    - **Aerial Camera**
    - **None**

    - **Falling Rain- Occlusion Zones + SDF + Depth Buffer**
    - **Falling Rain- Occlusion Zones + SDF**
    - **Falling Rain- Occlusion Zones + Depth Buffer**
    - **Falling Rain- Occlusion Zones**
    - **Falling Rain- Depth Buffer**
    - **Falling Rain- SDF + Depth Buffer**
    - **Falling Rain- Aerial Camera + Depth Buffer**
        -
    - **Falling Rain - Particle System**
        - Created with Unity's Particle System
        - Uses your scenes colliders for occlusion
    - **None**
        - Select if you don't want falling rain

Falling Rain VFX- Aerial Camera Occlusion

Falling Rain Camera:

1. Set the Culling Mask to only see layers that you want to occlude (mask out) rain
    a. Example... Default and Terrain
2. Set Volume Mask to Nothing

Render Texture:

1. Adjust x and y size to meet visual and performance needs (The default value is the recommended value)
    a. The smaller size the better
    b. X and Y need to be the same value
    c. If it's too small then the occlusion won't be as accurate
2. Leave all other settings as is, as this will result in the best performance and visual

VFX Property Binder

- Camera Binder for rain occlusion
- Transform Binder for DYNAMIC- Wind

Falling Rain VFX Particle System

★ **Max Particles and Emission Rat over Time are set via the DYNAMIC- RainWetness script's RainVFX Particle System**
★ **Tons of Settings in here to adjust but the main ones are...**

- **3D Start Size X and Y:**
    a. Adjust to set size of rain
- **Start Color:**
    a. Change color
- **Velocity Over Lifetime:**
    a. Change Linear Y min and max range to change the down velocity of rain
    b. Change Linear Z min and max range to change the sideways velocity of rain
- **Color Over Lifetime:**
    a. Change color and alpha

- **Collisions:**
    a. Change Collides With to layers you want to occlude rain
- **Sub Emitter:**
    ★ **This is turned off by default for performance reasons, as the particles system is CPU heavy**
    a. Controls what particles spawned when the rain particles collide with a collider

DYNAMIC: Multi Tag

This holds all of the manager scripts for the RainWetness systems. Which allows you to access a wide range of settings from 1 game object.

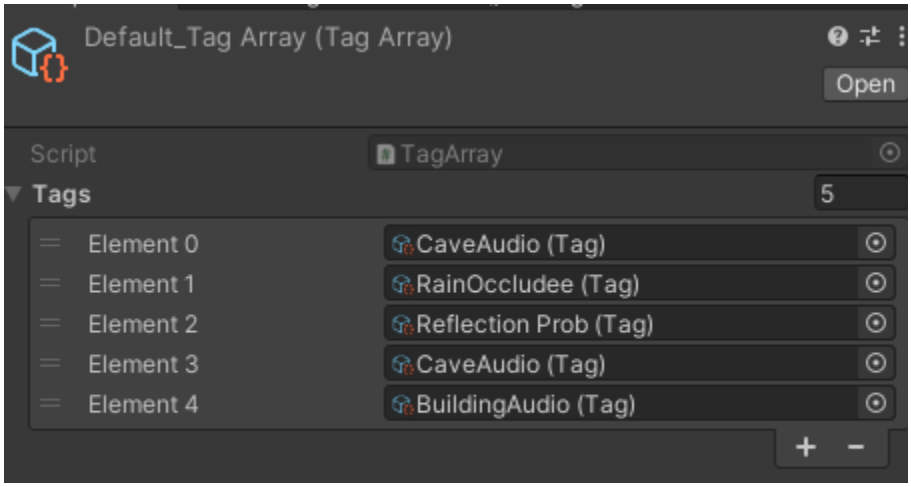
Tag Arrays

Create:

- Go to...Tools/MoveRotateResize/DYNAMIC- Multi Tag/Tag Array

Setup:

- Press + to add Tags to array
- Press - to remove a Tag from array



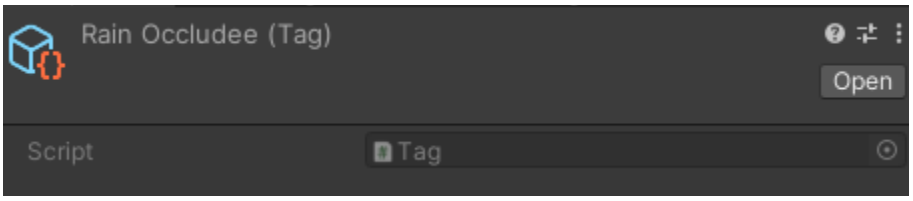
Tags

Create

- Go to...Tools/MoveRotateResize/DYNAMIC- Multi Tag/Tag

Setup

- Change name to desired tag name



Setting up Gameobject to Work with DYNAMIC: Multi Tag

Adding the Dynamic Multi Tag script

1. Select your gameobject
2. Click Add Component
    a. Add Dynamic Multi Tag Script
3. Press ADD beside the tag(s) you want to add to the gameobject



Scripting Example

Trigger Activated by Tag

4. Add to top of script

```
using MoveRotateResize.Dynamic.MultiTag;
```

5. Add a tag variable to your script

```
[SerializeField] private Tag RainOccludeeTag = null;
```

6. Example of trigger that checks for gameobject with tag

```
public void OnTriggerEnter(Collider other)
{
      // If object entering trigger doesn't have the tag then return
      if (!other.HasTag(RainOccludeeTag))
      Return;

      Debug.Log("Object with valid Tag entered trigger: ");
      // Put the rest of your code here...
```

```
}
```

Demos

Downloadable Demo

- Go to...DYNAMIC: Rain/Wetness by MoveRotateResize

Editor Demos

- Each demo shows a feature

(Enviro 3 Integration Demo)

- Go to the yellow box to make it sunny or the blue box to make it rainy. This showcases how DYNAMIC: Rain/Wetness is integrated to work with Enviro 3. As effects from DYNAMIC: Rain/Wetness update with the weather presets from Enviro 3

★ **Requires you to have Enviro- Sky and Weather installed**

1. Go to... Tools/MoveRotateResizes/DYNAMIC- RainWetness/Demos/Enviro 3 - Sky and Weather Integration

RainWetness Shaders

Default

Adds rain effects like wetness, rain streaks, and raindrops that will dynamically cover your game objects. Rain streaks and drops are dynamically placed on game objects based on gravity like settings. In other words this means that the raindrops appear on the top and sides of objects, and rain streaks appear only on the sides of objects. You can also control the porousness of game objects; which lets you specify if a game object is nonporous or porous. The latter will result in the game object appearing more wet with brighter colours. As an example, think of how rocks look when it's raining; they appear more wet and their colour gets more saturated. This effect can be controlled via sliders, Enviro- Sky and Weather Integration, or via the Dynamic Weather Rain script.



Surface Options

Surface Type • Opaque • Transparent	**Rendering Pass** • Default	**Cull Mode** • Back • Front
Check Boxes • Sorting Priority • Receive Fog • Alpha Clipping • Transparent Depth Prepass • Transparent Writes Motion Vectors • Depth Write	**Double-Sided GI** • Auto • On • Off	**Normal Mode** • Flip • Mirror • None

• Receive Decals • Double-Sided • Receive SSR Transparent		

Main Properties

Tiling/Offset • X = X Tiling • Y = Y Tiling • Z = X Offset • W= Y Offset	**Colour Picker** • Color is multiplied on top of the base color texture • Alpha Channel Control the overall visibility of the material • You can attach the "Dynamic Random Colour Per Object" script to a gameobject to make the colour picker choose a random color to assign it from an array. This effect only shows in play mode ■ Example... Auto assign random colour variations to all rocks in the scene	**Random Color Per Object** • Changes the Color Pickers color per object • Example... Makes it so each grass game object placed in the world have a slightly different color **Only use for stationary object, as color changes depending on the object's position **Use the script "Dynamic Random Colour Per Object'' to do the same thing. But that script lets you move the object without the color changing (as the script is not based on world position like the shader toggle)
Alpha Clip Threshold • Changes the threshold to for clipping	**Base Colour Map (RGBA)** • R= Reds • G= Greens • B= Blues • A= Alpha Mask (example... tree leaves)	**Smoothness View Angle Offset** • Object

Normal Map Properties

Normal Map Space • Object • Tangent	**Normal Map** (tangent space normals) **Normal OS Map** (object space normals)	**Normal Strength**

Mask Map Properties

Mask Map (MADS) • R= Metallic • G= AO • B= Detail Mask • A= Smoothness	**Sliders** • Metallic Remap Min • Metallic Remap Max • AO Remap Min • AO Remap Max • Detail Mask Remap Min • Detail Mask Remap Max • Smoothness Remap Min • Smoothness Remap Max

Detail Map Properties

Detail Map	Channels	Sliders
• Detail Tiling/Offset ○ X = X Tiling ○ Y = Y Tiling ○ Z = X Offset ○ W= Y Offset	• R= Desaturated Albedo • G= Normals Y • B= Smoothness • A= Normal X	• Detail Albedo Strength • Detail Smoothness Strength • Detail Normal Strength

Effects Mask Map Properties

Mask out the RainWetness effects with the R channel. Lets you use the combination of a texture and vertex painting to mask out maskExample... make the inside of a building/wall not have rain but still have the outside have rain. Recommend using Substance painter to paint this mask

Effects Mask Map
- R= Rain Mask
- G= *Not Used
- B= *Not Used
- A= *Not Used

Vertex Painting

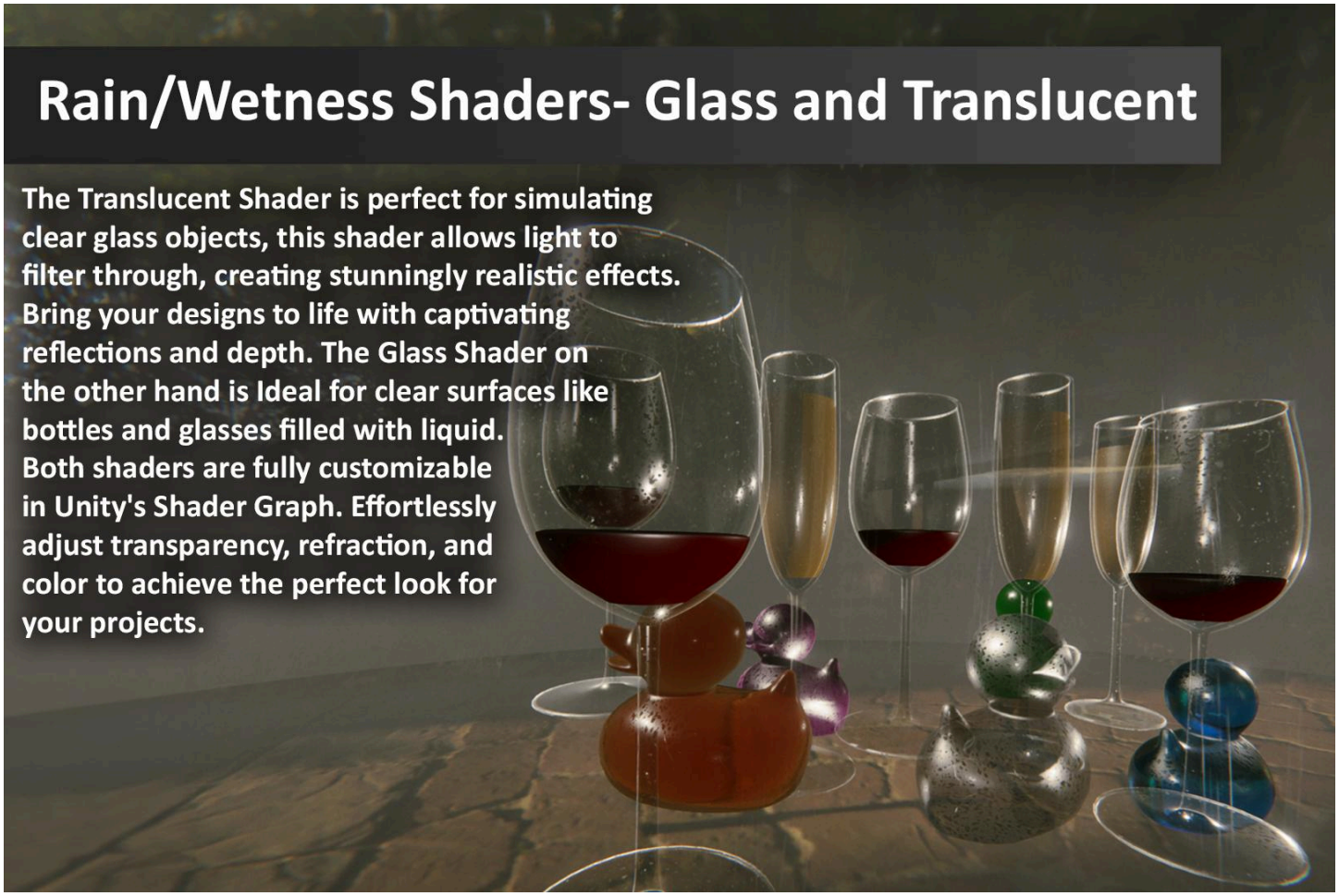
Mask out the RainWetness effects with the R channel. Example... make the inside of a building/wall not have rain but still have the outside have rain. Recommend using Unity's Poly Brush add-on to paint this mask in editor

Vertex Painting
- R= Rain Mask
- G= *Not Used
- B= *Not Used
- A= *Not Used

Advanced Options

Toggles/Dropdowns
- Enable GPU Instancing
- Global Illumination Dropdown
  - Realtime Lighting
  - Baked
  - None
- Enable Motion Vector for Vertex Animation
- Add Precomputed Velocity

Translucent/Glass



Surface Options (Same as Default Shader + the Following)

Surface Type • Opaque • Transparent	**Rendering Pass** • Before Refraction • Default • Low Resolution	**Blending Mode** • Alpha • Additive • Premultiply
Check Boxes • Sorting Priority • Receive Fog • Transparent Depth Prepass • Transparent Writes Motion Vectors • Depth Write • Receive Decals • Receive SSR Transparent	**Depth Test** • Disabled • Never • Less • Equal • LessEqual • Greater • NotEqual • GreaterEqual • Always	**Normal Mode** • Flip • Mirror • None
Double-Sided GI • Auto • On • Off		

Translucent/Glass Settings

Color Alpha
- Controls transparency of glass

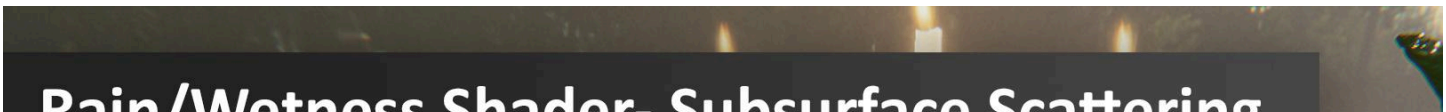
Translucent Settings

Diffusion Profile
Index of Refraction
Thickness
Transmittance Color
Transmittance Absorption Distance

Glass Settings

Fresnel Toggle
Fresnel

Subsurface



Same Features as the DYNAMIC- RainWetness- Default

Subsurface Scattering Settings

- **Diffusion Profile**
- **RGB Mask**
    - Sub Surface
    - Transmission
    - Thickness Mask
- **Subsurface Mask Remap**
- **Transmission Mask Remap**
- **Thickness Strength**

Anisotropy

Bring surfaces to life with sleek, directional highlights and realistic reflections. Whether it’s the brushed metal of a high-tech gadget, the polished surface of a car, or the smooth texture of a horse's fur, this shader adds a unique, polished touch to any project. Built entirely in Unity’s Shader Graph, the anisotropy shader offers full customization,allowing you to control the intensity of highlights with ease.

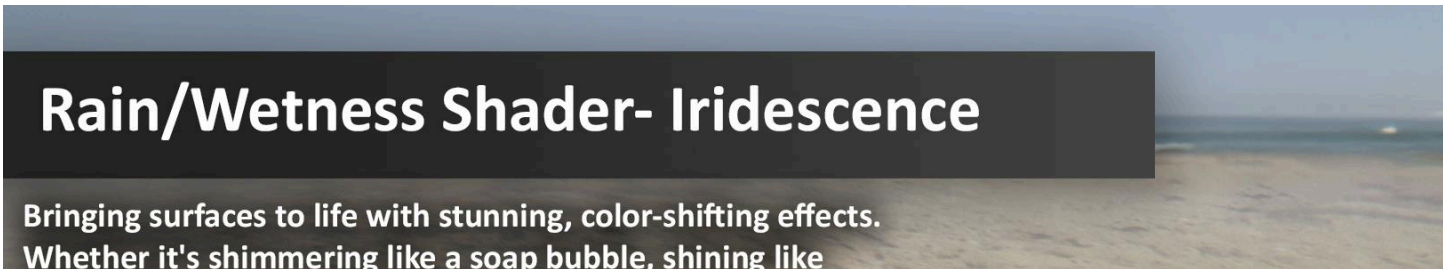


Same Features as the DYNAMIC- RainWetness- Default

Anisotropy Settings

- Use Texture
- Tangent Map
- Anisotropy Strength

Iridescence



Same Features as the DYNAMIC- RainWetness- Default

Surface Options (Same as Default Shader + the Following)

- **Surface Type**
  - Opaque
  - Transparent

Iridescence Settings

• **Index of Refraction** • **Thickness** • **Transmittance Color** • **Transmittance Absorption Distance**	• **Iridescence Map** ○ R= Mask ○ G= Thickness	• **Iridescence Mask Remap** ○ X ○ Y • **Iridescence Thickness Remap** ○ X ○ Y

Smooth Tessellation (each shader has its own version)



Same Features as the DYNAMIC- RainWetness- Default

Tessellation Settings

- Makes objects appear smoother by increasing the poly count of your objects mesh using tessellation
- **Currently DOESN'T include height displacement only mesh smoothing, which subdivides the mesh when the player is close

Integrations

Enviro 3 - Sky and Weather

Experience full synchronization between Enviro 3's advanced weather system and our cutting-edge rain and wetness effects. This direct integration ensures smooth and immersive transitions, perfectly aligning visual and environmental effects with changing weather conditions.



Seamless Integration

- Experience full synchronization between Enviro 3's advanced weather system and our cutting-edge rain and wetness effects. This direct integration ensures smooth and immersive transitions, perfectly aligning visual and environmental effects with changing weather conditions.

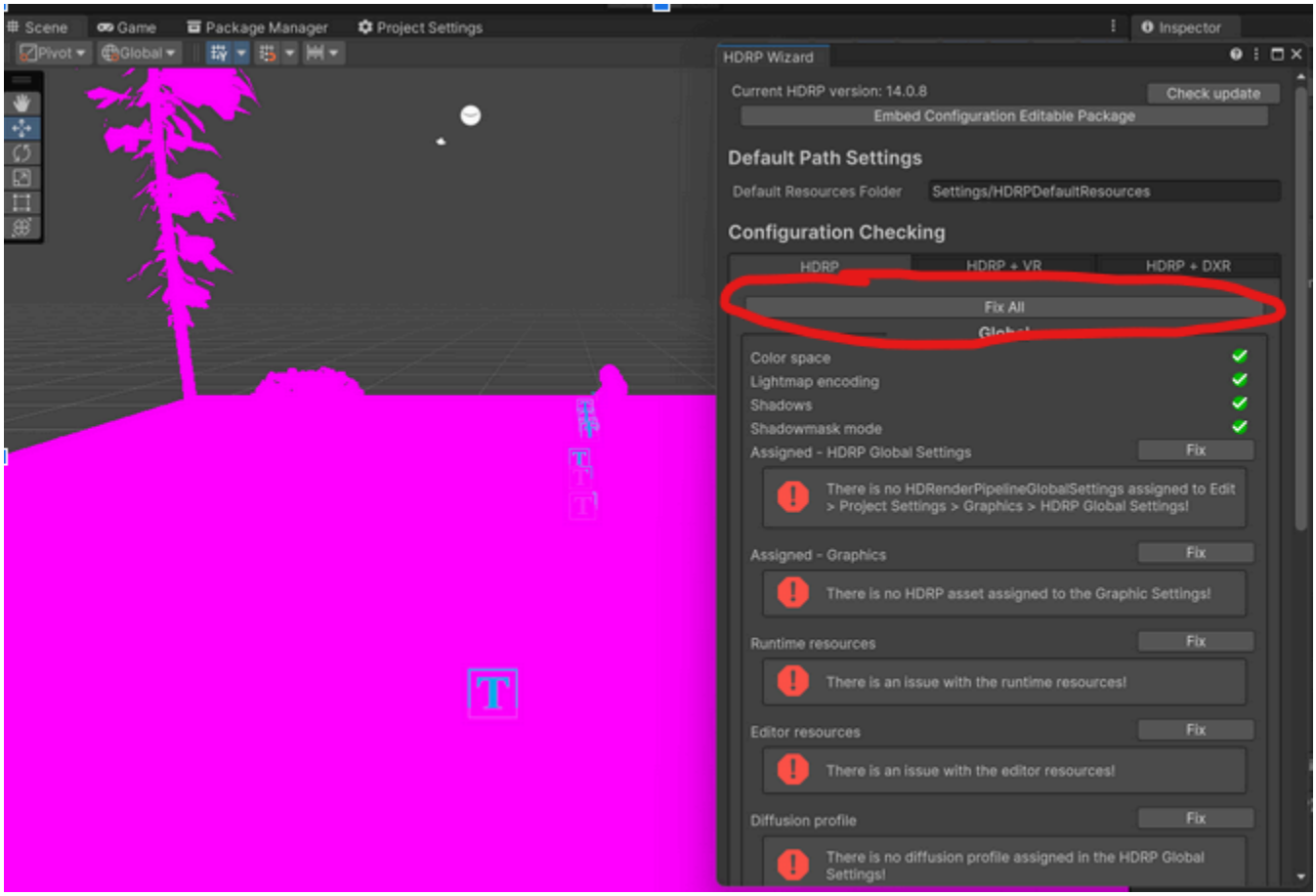
Interactive Demo Showcases

- **Yellow Enviro Zone:** Activate the Sunny Weather Preset and enjoy a bright, clear environment.
- **Blue Enviro Zone:** Switch to the Rainy Weather Preset and watch as rain transforms the world around you in real time.

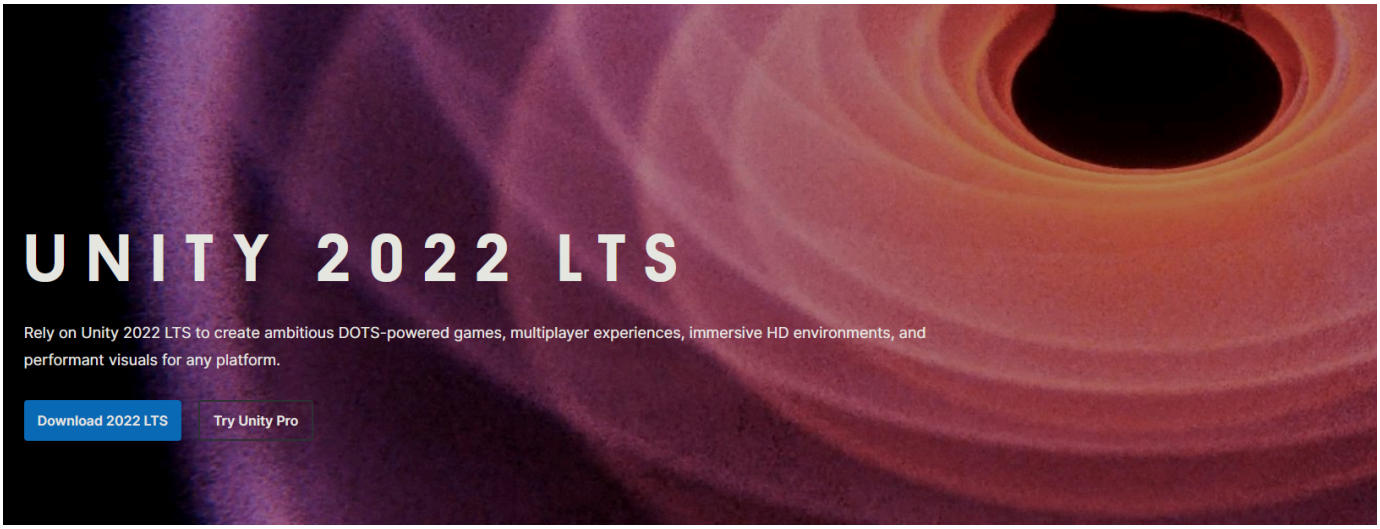
These demos highlight the flawless integration of Enviro 3 with our weather effects, delivering unparalleled realism and atmosphere to your Unity projects.

FAQ

Problem: Textures are Pink	**Solution:** Go to Windows 1. Rendering/HDRP Wizard a. Click “Fix All”

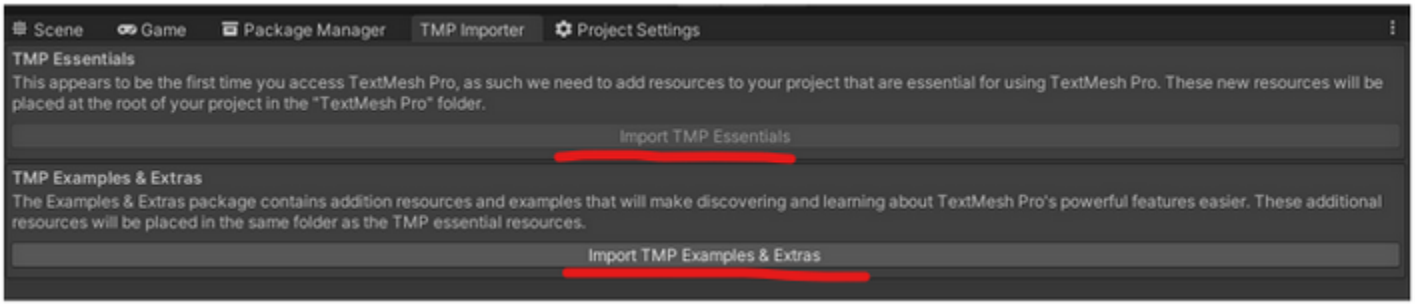


Problem: Raindrops aren't visible or scaling correctly	**Solution:** This means that you are using a Unity version lower than 2022.2+. As Unity Version 2022.2+ is the minimum requirement for this assest to work correctly.

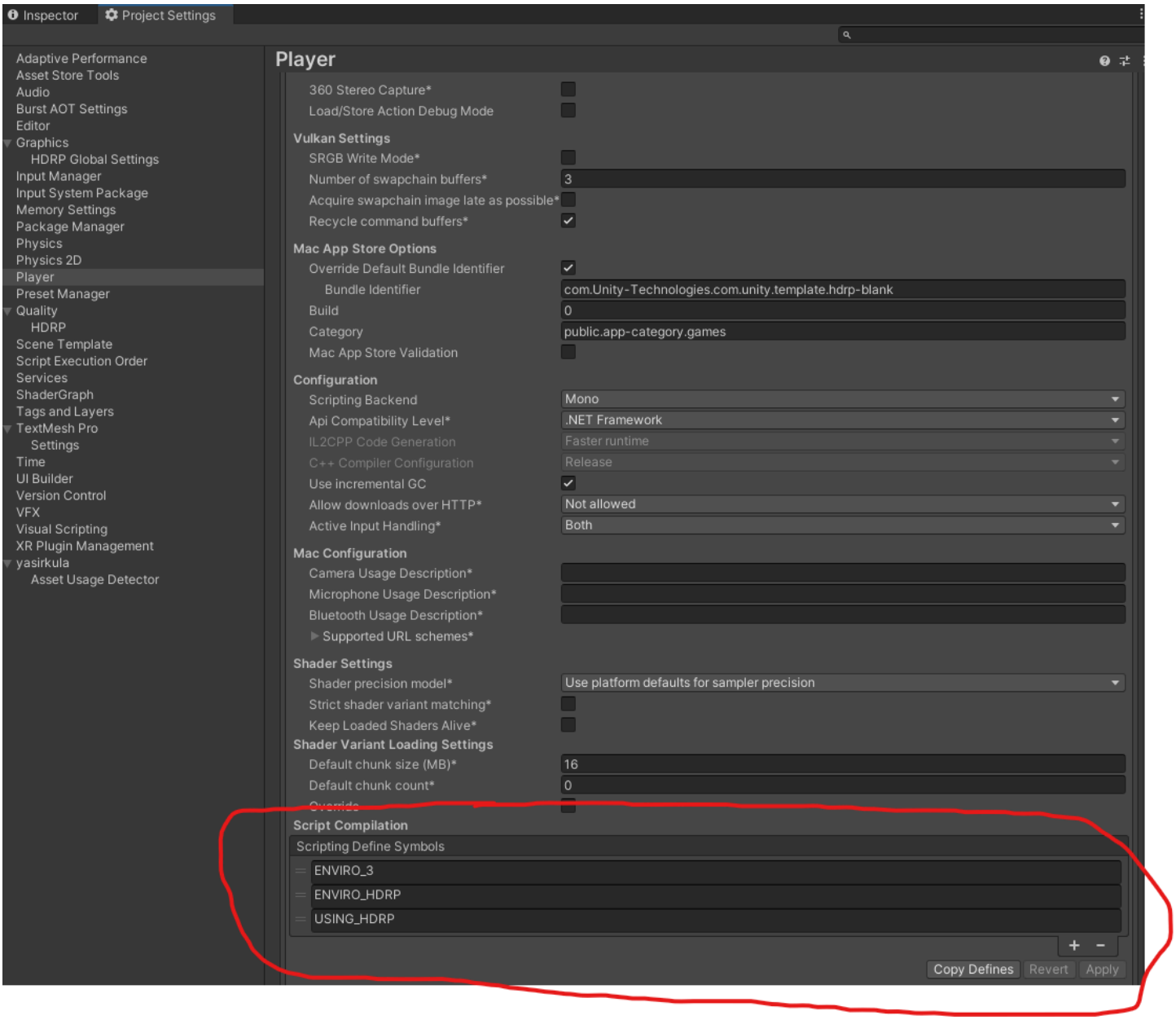


Problem:	**Solution:**
Getting Errors for TextMesh Pro	Go to and click... Tools/MoveRotateResize/Setup

	i. FirstTimeSetup Or... • Manually make sure you have the same settings as the red highlighted areas



Problem: Getting Errors for missing definitions (example... #USING_HDRP or #USING_URP). *Your project CAN'T have Using HDRP and Using URR at the sametime. Please only use one	**Solution:** Go to and click... Tools/MoveRotateResize/First Time Setup Or... • Manually make sure you have the same settings as the red highlighted areas



Special Thanks

Thanks to my beautiful wife for all of her love and support!!

If you need support, questions, or have any suggestions please email me at MoveRotateResize@gmail.com

© MoveRotateResize

British Columbia, Canada

Changelog/Patch Notes

Legend

- = Please Stop and Read
- = Working On
- = Completed

Version 1.3.2 (NOT Released Yet)

1. **Miscellaneous Error Fixes**

★ **!!STOP!! Process for updating to new version**

1. Before updating to the new version please make sure to make a copy of your ENTIRE MoveRotateResize folder and paste it to your desktop or another location
2. Once that is completed please delete the entire MoveRotateResize folder that's inside of your Unity project
3. Now you can update and install new version of the asset

- **All Shaders**
  i. Changed _Global_Rain_Occlusion variable name to Rain_Occlusion to align with this assets naming conventions

- **DYNAMIC: Systems Manager script**
  i. Updated debug log wording for player and multi. player detection. To let the user know that only 1 gameobject can be tagged player in a scene at a time
  ii. Changed when the DYNAMIC: Falling Rain variable gets assigned. As it was too low in the execution order and was causing null errors

- **MoveRotateResize First Time Setup script**
  i. Updated function name that executes the MoveRotateResize Input Setup script
  ii. Added 2 new script definitions called USING_UNITY_6 and USING_UNIT_2022. These are used with the new Puddle Decal Layer script to help it know what Unity Version you are using

- **Puddle Decal Layer script**
  i. Sets the decal layer(s) of your puddle and wetness decals automatically. As different Unity versions have different code that sets decal layers

- **MoveRotateResize Input Setup script**
  i. Added missing code to auto add Right Stick X and Y to the Old Input System

- **DYNAMIC: Occlusion script**
  i. Changed _Global_Rain_Occlusion variable name to Rain_Occlusion to align with this assets naming conventions
  ii. Added debugShowEditModeUpdateWarning bool to let you disable this warning
  iii. Updated how Edit Mode Occlusion is disabled when going in and out of playmode'
  iv. Removed SDF Occlusion setup in ONValidate() method, as it was causing a null ref error

- **Occlusion Zone script**
  i. Added OnValidate() method to get the Zone Collider variable in edit mode

- **DYNAMIC: Falling Rain script**
  i. Changed [DefaultExecutionOrder(-100)] to [DefaultExecutionOrder(-500)] to fix errors in the DYNAMIC: Occlusion script

- **DYNAMIC: Tools Find Instanced Materials script**
  i. Refactored code to work more reliably
  ii. Removed the ability to search all project scenes. As this wasn't working correctly

- **Multiple Scripts**
  i. Removed some debug logs that weren't needed anymore
  ii. Removed some unused variables

- **Demos**
  i. Added missing Event Systems to some scenes
  ii. Removed some missing scripts from some gameobjects
  iii. Replaced Default input system object to one from the shared folder
  iv. Added missing descriptions to each demo scene

- **Manual**
  i. Added missing Install Instructions to enable the Old Input System

Version 1.3.1

1. **Misc. Fixes and Optimizations**
2. **Added variables for Rain/Wetness' World Space mask**
3. **Fixed Normal Blending of Rain Drops/Streaks**
4. **Fixed Dripping Rain Spawn Point Issues**
5. **Fixed Post Process Volumes to prevent them becoming Instances**

★ **!!STOP!! Process for updating to new version (if updating from version 1.3)**

1. Simply download and install hotfix. You DON'T need to delete your folder this time.
2. Go to Tool/MoveRotateResize/ DYNAMIC- RainWetness/First Time Setup/Setup. Press skip on the shader install.

★ **!!STOP!! Process for updating to new version (if updating from version 1.2 or lower)**

1. Before updating to the new version please make sure to make a copy of your ENTIRE MoveRotateResize folder and paste it to your desktop or another location
2. Once that is completed please delete the entire MoveRotateResize folder that's inside of your Unity project
3. Now you can update and install new version of the asset

- **VFX Dripping Rain_01**
  i. Changed Turbulence mode to absolute to help reduce the particles "exploding" when placed a spawn points
  ii. Updated gravity to use World Space so particles always go towards the ground (-Y)

- **Spawn Dripping Rain script**
  i. Added missing code to allow VFX's scale to be updated at runtime
  ii. Added missing code to update the VFX's Global Rain Extended value
  iii. Fixed error where edit mode spawns wouldn't update their position, rotation, or scale after entering/exiting playmode. This was done by changing a few variables to [SerializeField, HideInInspector]

- **SPAWN POINT- Interior Audio prefab**
  i. Added missing Spawn Point- Audio tag that DYNAMIC:Smart Spawner script requires in order to spawn these objects
  ii. Increased Box Collider size to 30 for all axis' to make these spawns be able to spawn farther away from the player

- **Falling Rain VFX- Camera Occlusion**
  i. Removed this prefab and all linked VFX graphs. As all of these settings and features have been combined into the falling rain VFX Graph prefab

- **Falling Rain VFX prefab**
  i. Increased default min/max velocity of rain value
  ii. Updated multiple default settings to increase rain shininess in sunlight/light sources
  iii. Added all different occlusion features to 1 prefab. This will allow speeder updates from us, as less prefabs have to be managed

- **Falling Rain VFX graph and textures**
  i. Fixed it so rain now shimmers in light. As the angle was set to the wrong value in the VFX graph
  ii. Rotated falling rain's base color and normal textures 180 degrees

- **Global Post Processing prefab**
  i. Fixed post processing volumes from becoming instances by changing how their settings are updated in scripts using the shared volume profile and overriding settings using .Override()

- **DYNAMIC: Rain Wetness script**
  i. Fixed post processing volumes from becoming instances by changing how their settings are updated in scripts using the shared volume profile and overriding settings using .Override()
  ii. Optimized the UpdateFog() method to increase performance

- **DYNAMIC: Falling Rain script**
  i. Moved the "Enable SDF Occlusion" bool to the DYNAMIC: Falling Rain script and make it hidden, as this is now handled automatically when selecting your falling rain type using the dropdown
  ii. Updated it so when selecting your falling rain type using the dropdown it updates bools in the background to enable/disable features

- **DYNAMIC: Smart Spawning script**
  i. Updated script to only perform VFX Occlusion functions if the falling rain type enables that feature
  ii. Added a new variable called maxSpawnAndRemovalDistance that works with the Camera Frustum mode to allow spawned VFX to stay active when near the player. As without this VFX above the player would disappear when out of the camera's frustum

- **DYNAMIC: Occlusion script**
  i. Updated script to only perform SDF Occlusion functions if the falling rain type enables that feature

- **DYNAMIC: Light Flicker script**
  i. Added a new script that lets you add flickering to your HDRP lights

- **DYNAMIC: Systems Manager script**
  i. Updated debug logs to only tell you if components are missing

- **Default Systems sub-graph**
  i. Created a new sub-shader that holds all default sub-graphs. This will help make updating shaders easier as they'll all share a common sub-shader

- **Detail and Rain Wetness sub-graphs**
  i. Moved the detail normal map combining to the rain wetness subgraph and added a new lerp for all rain effects. As the detail map normals were overriding the normals of the rain drops and rain streaks

- **Absolute World Space Mask sub-graph**
  i. Updated to expose 2 variables that allow the user to adjust the mask's position and strength. This will help you save time so you don't have to create masks or vertex paint every object to mask out the rain on the bottom of objects. Example... your roof asset will automatically appear wet on the top,and the underside will be completely dry

- **Detail Map sub-graph**
  i. Removed the enable detail mask enum and made the system work automatically. Please change the min and max of the mask maps detail mask remaps to 0 if you want to hide the effects of the mask map
  ii. Removed the normal map input and output. As this is now combined in the RainWetness subgraph
  iii. Remove some node

- **Rain Wetness sub-graph**
  i. Added 2 variables that control the Absolute World Space Rain mask. This allows you to make adjustments per material to specify what counts as the bottom of a mesh
  ii. Fixed how the metal mask was applied as it was using a 1 - by mistake
  iii. Fixed it so that the detail mask regular smoothness and rain smoothness all work together by reconfiguring some nodes
  iv. Removed the metal mask as it's no longer needed. This fixes the porous/non-porous slider functionality

- **RainWetness Systems (Copy Me) sub-graph**
  i. Updated to include all new features and setups

- **DYNAMIC: Falling Rain (Day) demo**
  i. Added new demo to showcase falling rain in the day time

- **DYNAMIC: Falling Rain (Night) demo**
  i. Added new demo to showcase falling rain at night

- **Naughty Attributes**
  i. Deleted Test folder. As it's no longer needed

- **Occlusion Zone Material**
  i. Changed Rendering Pass to "Low Resolution". This fixes the issue where transparent VFX and objects were being hidden when viewed from the other side of an Occlusion Zone

- **Fix Missing Input Axis'**
  i. Added a new script to auto add our d-pad inputs. Aftering updating to this version please go to Tool/MoveRotateResize/ DYNAMIC- RainWetness/First Time Set Up/Set Up. You can keep the shader install, as you just need the first time setup script to add the missing keybinds

- **Occlusion Zone script**
  i. Commented out some debug logs that aren't needed

Version 1.3

1. **Added DYNAMIC: Dripping Rain**
2. **Easily place and customize dripping rain zones in your scenes**
3. **Added DYNAMIC: Smart Spawn**
4. **Object Pooling for dripping rain and interior rain audio**
5. **Added Ray Tracing Compatibility for all shaders**
6. **Improved DYNAMIC: Occlusion (for Shaders and Falling Rain VFX)**
7. **Added VFX Occlusion Zones**
8. **Added SDF Occlusion Zones**
9. **Improved Terrain Puddle Shader**
10. **Optimizations and Misc Fixes**

★ !!STOP!! Process for updating to new version

Version 1.3 is a big update and changes/optimizes many systems. As a result you'll have to delete your old DYNAMIC- Rain Wetness Systems gameobject in your scenes and add a new one. Sorry for the inconvenience.

1. Before updating to the new version please make sure to make a copy of your ENTIRE MoveRotateResize folder and paste it to your desktop or another location
2. Once that is completed please delete the entire MoveRotateResize folder that's inside of your Unity project
3. Now you can update and install new version of the asset

★ !!STOP!! If you used the puddle painter

1. The "rt_puddle_and_wetness" render texture was increased from 1024 x 1024 to 2048 x 2048 resolution. As a result you will have to either delete your Puddle Paint Mask texture for each of your custom painted terrain or change the "rt_puddle_and_wetness" render texture back to 1024 x 1024

- **The DYNAMIC- Rain/Wetness Systems prefab**
  i. Updated to be the new home for all main manager scripts. As before it only held the Do Not Destroy On Load script
  ii. Added the DYNAMIC: Wind script
  iii. Added the DYNAMIC: Reflection Probe script
  iv. Added Wind Zone gameobject

v. Add DYNAMIC: Falling Rain script
vi. Added Reflection Probe- Spawner gameobject
vii. Added Reflection Probe- Global gameobject
viii. Removed/Deleted the Do Not Destroy On Load script. As this has now been integrated into the DYNAMIC: Systems Manager script

- **All Shaders**
  i. Refactored all shaders to be compatible with Ray Tracing. As before all gameobjects using our shaders would look black when ray tracing was enabled
  ii. Changed default values for Streak Normal Strength and Steak Normal Distortion to 1 for all shaders
  iii. Fixed some variables that had the wrong names. This fixes errors on some shaders where global variables weren't being updated when changing them in the DYNAMIC: Rain Wetness script
  iv. Added an absolute node before all A's and B's of power nodes to fix make it so power can't take a negative number as this causes build error logs

- **All Scripts**
  i. Refactored how Global Variables are get/set

- **Demos**
  i. Created new demo to showcase DYNAMIC: Dripping Rain features
  ii. Renamed Full Features demo to Showcase- DYNAMIC: Rain Audio

- **Rain/Wetness sub-shader**
  i. Replaced the raindrop texture with a new one that has more raindrops
  ii. Refactored know raindrops and rain streaks are created in the shader
  iii. Increased default resolution of raindrops and rain streak textures
  iv. Cleaned up and organized the layout of nodes

- **Terrain Puddle shader**
  i. Refactored and cleaned up entire shader
  ii. Updated ripples with a new texture
  iii. Added waves to puddles
  iv. Refactored nodes that are attached to the Puddle Mask Remap variable. As it was causing puddles not to show when Global Rain Extended value was lower than 0.5

- **Misc Subgraphs**
  i. Added new subgraph called UV's- Hex Sample Texture 2D
  ii. Changed some subgraphs naming conventions

- **DYNAMIC: Puddle Painter**
  i. Changed rt_puddle_and_wetness size to 2048 x 2048. As the painted layers looked pixelated at 1024 x 1024
  ii. Updated it so when creating a new puddle group the system will auto-save the new spawned prefab changes. As before when opening the newly created prefab it would still link to things form the default folder

- **DYNAMIC: Wind**
  i. Updated the Falling Rain VFX wind direction to work in 360 degrees
  ii.
  iii. Removed the DYNAMIC: Wind script from the gameobject and moved it to the DYNAMIC: Rain/Wetness Systems prefab
  iv. Changed random rotation to only work in playmode. As it was causing all local fog volumes to jitter in edit mode
  v. Refactored to increase performance
  vi. Changed the windZone variable to a SerializableField so it can be assigned in the inspector
  vii. Updated inspector layout
  viii. Added info boxes to inspector

- **DYNAMIC: Dripping Rain**
    i. Created a new set of scripts that controls the spawning of dripping rain VFX. These can be placed on static or movable objects

- **DYNAMIC: Rain Wetness**
    i. Refactored to increase performance
    ii. Organized inspector
    iii. Moved all Falling Rain VFX code to new script call DYNAMIC: Falling Rain
    iv. Renamed Rain Streak Distortion Strength to Rain Streak Distortion and changed default value to 0.05. Please update this value in your scene if required, as changing variable names resets the value you manually entered
    v. Fixed Rain Toggle so it'll update to the correct value even in the middle of transitioning the Global Rain and Global Rain Extended values

- **DYNAMIC: Falling Rain**
    i. Update and fixed how the wind direction is calculated
    ii. Moved all Falling Rain VFX code from the DYNAMIC: Rain Wetness script to this new script
    iii. Changed some variable names
    iv. Removed some variables
    v. Made it so the particle spawn amount is controlled from the VFX Graph (except the particle system which is still controlled from the script)
    vi. Added a dropdown that lets you select your falling rain type. This dropdown is auto populated with all gameobjects that are children of the Falling Rain VFXs gameobject
    vii. Replaced the DYNAMIC: Follower script on all Falling Rain VFX gameobjects with the new DYNAMIC: Falling Rain Player Follower script.
        1. This new script makes the Falling Rain VFX always be in front and above the player. Ensuring that the falling rain is only where the player can see which increased performance

VFX Ground Fog shader

i. Update and fixed how the wind direction is calculated
ii. Update look and feel
iii. Add Hex Tiling to help remove tiling repetition

- **DYNAMIC: Rain Audio**
    i. Separated all sphere cast logic to a new script called "DYNAMIC: Smart Spawn". This allows easier expandability
    ii. Refactored script to increase performance

- **DYNAMIC: Smart Spawning**
    i. Separated the sphere cast code from the Dynamic: Rain Audio script to create this new script
    ii. Refactored original code to make compatible with multi scripts, added new features, and increased performance
    iii. Updated to work with spawn points/spawns from DYNAMIC: Rain Audio, DYNAMIC: Dripping Rain, and VFX Occlusion Zones from the DYNAMIC: Occlusion script
    iv. Added new tags for VFX Occlusion Zone, Occlusion Zone, and Audio Zone

- **DYNAMIC: Occlusion**
    i. Added an array called VFX Occlusion Spawns. This is used for the updated VFX Occlusion system

- **DYNAMIC: Occlusion Zones**
    i. Added new tag called Occlusion Zone
    ii. Added Dynamic: Multi Tag script and set the tag to Occlusion Zone. This is used to spawn the new VFX Occlusion Spawns
    iii. Removed all Pro- Builder Occlusion Zones, as the new occlusion system requires zones to be boxes

- **DYNAMIC: Reflection Probes and DYNAMIC: Reflection Probes Spawner**
    i. Refactored scripts
    ii. Changed some variable names
    iii. Organized inspector
    iv. Removed editor script
    v. Removed debugRayYOffset variable and updated script to use the spawnYOffset instead

- **DYNAMIC: Systems Manager**
    i. Renamed Global Variables Manager to DYNAMIC: Systems Manager
    ii. Refactored how Global Variables are set up
    iii. Added do not destroy on load checkbox
    iv. Added buttons to enable/disable debug Gizmos for all scripts

- **DYNAMIC: Controller First Person**
    i. Updated to work with controllers (currently only works in playable demo)

- **DYNAMIC: Fly Camera**
    i. Updated to work with controllers (currently only works in playable demo)

- **Do Not Destroy script**
    i. Removed this script and added its functionality to the DYNAMIC: Systems Manager script

- **Move Rotate Resize Demos script**
    i. Added a warning to save your scene before loading a demo
    ii. Added shortcut to the Dripping Rain Demo
    iii. Moved Demos tab to the main tab

- **Move Rotate Resize Tools script**
    i. Updated tool option to spawn the new version of the DYNAMIC: Rain/Wetness Systems prefab

- **Falling Rain VFX**
    ★ **Please be advised that you might have to adjust some settings to get the old look you had. As this VFX Graph has gone through a minor upgrade, and there now is multiple different version to choose from**
    i. Changed Falling Rain VFX gameobject names as there care multi different ones now
    ii. Refactored how rain splashes are spawned
    iii. Refactored VFX Graph to work with the new VFX collision system
    iv. Changed all gradients colors to color and alpha. This allows you to have select the rain color you want easier
    v. Removed the set alpha particle fade variable. Now this settings is controlled via the Global Rain variable
    vi. Updated it so the particle amount goes down and up with the Global Rain amount instead of using set alpha particle fade (which was removed)
    vii. Added rain splash min/max size variables
    viii. Added rain splash smoothness variable
    ix. Changed some default variable values
    x. Renamed some variables

- **Mark Up Attributes (Inspector Organization)**
    i. Added this 3rd party asset to help organize this assets inspector windows
    ii. Refactored/Organized all scripts with this new asset
    iii. Added custom editors for all scripts that use this new asset

- **Mesh Combiner**
    i. Added this 3rd party asset to help the user to easily create SDF files for falling rain occlusion

- **Misc Textures**
    i. Reduced some texture file sizes to help reduce overall package size

ii. Deleted some unused assets to help reduce overall file size

- **Downloadable Demo**
  i. Revamped demo and added all new/updated scenes
  ii. Updated demo to work with both controllers, keyboard, and mouse

Version 1.2

- **Fixed Edit Mode Occlusion**
- **Added 2 Tools**
- **Increased Stability**
- **Updated some Demos**

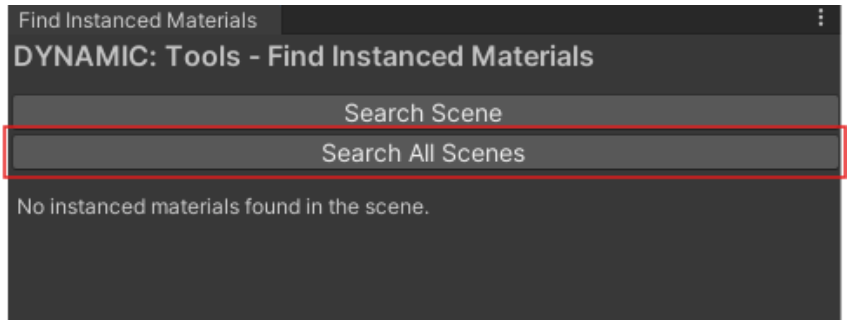
★ !!STOP!! Process for Updating to New Version

1. Before updating to the new version please make sure to make a copy of your ENTIRE MoveRotateResize folder and paste it to your desktop or another location
2. Once that is completed please delete the entire MoveRotateResize folder that's inside of your Unity project
3. Now you can update and install new version of the asset

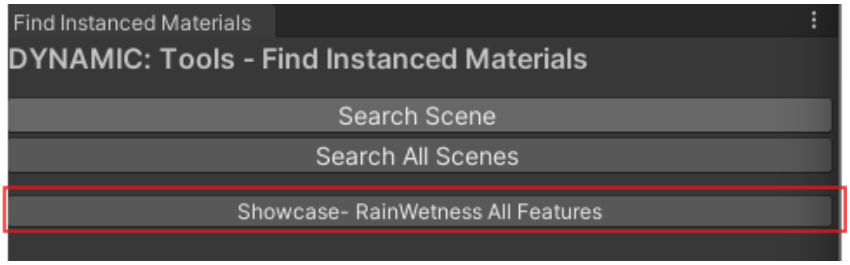
★ If you used "Edit Mode Occlusion" from version 1.1 or 1.1.a

An error in the previous version of Edit Mode Occlusion was causing gameobjects materials to permanently become instances, making the materials uneditable. The steps below fix this issue, and the new refactored version of the script prevents that from happening in the future. Very sorry for the inconvenience.

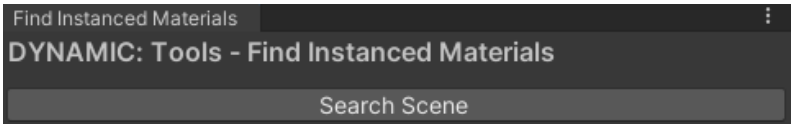
1. Go to each of your scenes that you used "Edit Mode Occlusion" and do the following...
2. Go to Tools/MoveRotateResize/DYNAMIC: Tools/Find Instanced Materials
3. Press Search All Scenes



1. A list of buttons for each scene that's affected will appear
2. Click the button for each scene to load it



3. Once a scene is loaded press "Search Scene" to check for instanced materials
4. A list of materials that are instanced will appear
   1. The system tries to auto find the original material, if it can't it'll say "None" in the material slot
   2. If it says "None" then click on the gameobject to jump to it
   3. Then replace the material manually with the correct one from your project folder
5. If the system auto found an original material and it looks correct then press "Replace"
6. Rinse and repeat this process thill no more scenes with the affected instanced materials can be found



- **DYNAMIC: Tools (These are FREE assets that will be released as standalone assets in the future)**
    i. Added a new section on the Toolbar labeled “DYNAMIC: Tools”. This will be the new home for all tools like this in the future
    ii. Added a new tool called “Find Instanced Materials”
    iii. Added a new tool called “Find and Change Layers”

- **Store Package Version**
    i. This version includes the fixes from version 1.1.a. As it looks like the package for version 1.1a didn’t upload correctly, as Unity’s package manager says that it’s still version 1.1.

- **Gameobject Layers**
    i. Used our DYNAMIC: Tools- Find and Change Layers tool to move our layers to 30 and 31
    ii. Removed Terrain and Character layers

- **Global Variables Manager script**
    i. Added bool to track when initialization is finished
    ii. Added Main Camera variable

- **DYNAMIC: Occlusion script**
    i. Refracted how Edit Mode Occlusion works. As it was causing gameobjects materials to permanently become instances, making the materials uneditable
    ii. Updated inspector organization
    iii. Added a layer mask variable for the occlusion zones

- **Occlusion Zone script**
    i. Refactored How Edit Mode Occlusion works. As it was causing gameobjects materials to permanently become instances, making the materials uneditable
    ii. Refactored how Occlusion initializes. Please wait 1 second for the system to start
    iii. Added a layer mask variable

- **DYNAMIC: Mesh Mirror script**
    i. Updated script to the new version. As the old version was uploaded by mistake since version 1.0

- **DYNAMIC: Multi Tag scripts**
    i. Changed Default Execution Order to -1000000
    ii. AddTags() is now called in Awake and OnEnable to increase reliability
    iii. Added Debug Logs to Tag Helper

- **Dynamic Puddle Painter Script**
    i. Made the brush gameobject visible in the inspector

- **PDF Manual**
    i. Updated included PDF to the newest version

- **Falling Rain (VFX Graph)**
    i. Increased rt_rain_depth_buffer size from 32 x 32 to 64 x 64, as the occlusion wasn’t accurate enough

- **Enviro Demo**
    i. Fixed toolbar not opening the demo when clicked

- **All RainWetness Shaders / Dynamic- RainWetness scripts (global variables)**
    i. Fixed all Global Shaders variables which weren't working. As some shaders had vector2’s when it they should have been floats, had wrong variable names, or had exposed variables when they should have been unexposed (Global)

- **Showcase- RainWetness Shaders- Glass and Translucent**
    i. Unchecked Double Sided for all wine in the scene, as it was causing graphical errors
    ii. Updated the lighting and some shader settings to improve the image quality of the scene

- **Showcase- RainWetness Shaders on Skinned Mesh Renderers**
    i. Added new demo scene, as zombies weren't CC0 (free)

- **Showcase- RainWetness Shaders- Subsurface Scattering**
    i. Updated demo to include more items

- **Showcase- RainWetness Shaders- Glass and Translucent**
    i. Updated demo lighting

- **Showcase-DYNAMIC- Occlusion (Static Objects)**
    i. Added new demo to show effect

- **Showcase-DYNAMIC- Occlusion (Moving Objects)**
    i. Added new demo to show effect

- **Demos**
    i. Renamed some demos
    ii. Added diffusion profiles to global volumes

Version 1.1.a

- **Build Project Hotfix**

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    2. Once that is completed please delete the entire MoveRotateResize folder that's inside of your Unity project
    3. Now you can update and install new version of the asset

- **Building Unity Projects**
    i. Fixed more build errors that arose from all the changes in patch 1.1
1. Scripts affected DynamicMultiTag, GlobalVariablesManager, TagArray, and AddPrefabsFromSameFolderOnHierarchyEnter

- **Showcase- RainWetness Shaders on Rotating Objects**
    i. Assigned missing particle system in Dynamic- RainWetness script
    ii. Updated Lighting to show off duck better

Version 1.1

- **Edit Mode Occlusion**
- **Shader Swapper Update**
- **Misc Fixes**

    ★ **!!STOP!! Process for Updating to New Version**
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    2. Once that is completed please delete the entire MoveRotateResize folder that's inside of your Unity project
    3. Now you can update and install new version of the asset

- **Building Unity Projects**
    i. Fixed Build errors when user tried to make a build
1. Some scripts where missing “#if UNITY_EDITOR“ around editor only features
2. Some editor scripts weren’t in Editor folders

- **First Time Setup Script**
    i. Fixed error in Unity 6 Preview where the script couldn’t locate the shader package when installing shaders
1. This fix makes it so shaders in Unity 6 Preview have the correct Material Type as newer versions of Unity changed that setting

- **All Dynamic RainWetness shaders**
    i. The fix in the “First Time Setup Script” makes sure that each shader has the correct Material Type set. As newer versions of Unity changed how the Material Type is set up

- **Shader Swapper**
    i. Added a button that allows you to swap shaders for the currently selected gameobject(s)
    ii. Added a button to let you to set the original shader based on the currently selected gameobject

- **Shader Property Finder**
    i. Updated so the finders variables are based on the same variables as the shader property finder
1. This makes it easier to find and add variables as snow the shader swapper and shader property finder use the same variables and categories
    ii. Updated script so found variables are green and ones that aren't found in the shader property group are red
1. This allows you to find missing variables in your Shader Property Group easier
    iii. Added the ability to add a shader base on the currently selected game object
    iv. Added warnings and more debugs

- **Shader Property Groups**
    i. Updated the Shader Property Group script so it notifies the Shader Property Finder script if any changes were made to one of its scriptable objects

- **Dynamic Occlusion and Occlusion Zones scripts**
    i. Added a red debug sphere to visualize the sphere size
    ii. Renamed playerSphereCastRadius variable to occlusionSphereSize
    iii. Added a Toggle to Enable Edit Mode Rain Occlusion.

★ **The toggle for Enable Edit Mode Occlusion will be auto disabled in playmode**

    iv. Also added a yellow debug sphere to visualize what items are being affected by edit mode rain occlusion
    v. Added a global place to set every Occlusion Zones Material
    vi. Added an Occlusion Zone Material Alpha variable. This lets you control all occlusion zones visibility in the editor

- **1st Person Controller Prefab**
    i. Added missing Dynamic Multi Tag component and set to RainOccludee
1. Now the interior audio volume control will work correctly when the player enters/exists an occlusion zone
- **Dynamic Pooling Manager script**
    i. Changed script name to Dynamic Rain Audio
    ii. Fixed null references and setup errors by letting the script get and set its own player variable. As using UnityEngine.Pool and getting Instances from singletons has script execution order problems

- **Spawn Interior Audio script**
    i. Refactored script to fix null references. As using UnityEngine.Pool and getting Instances from singletons has script execution order problems

- **Dynamic Puddle Painter Script**
    i. Refactored to fix null reference errors with assigning the paint brush

- **Add Prefabs From Same Folder On Hierarchy Enter**
    i. Moved this folder to the Runtime folder. This fixes the error where the engine can't find the script
    ii. Removed the code that made this script execute too early

- **Colour Random Per Object script**
    i. Changed name to "RandomColourPerObject"
    ii. Moved script to it's own folder
    iii. Refactored script to make it work with nested gameobjects. This means if a gameobject has children those children will have the same random color assigned. This fixes errors where gameobjects with LODs would all have random colors

- **All Scripts**
    i. Fixed potential memory leaks in scripts by stopping coroutines in and OnDisable()
    ii. Refactored how some instances are setup to fix null references due to script initialization order

- **Dynamic Puddle Painter Script**
    i. Refactored to fix null reference errors with assigning the paint brush

- **Add Prefabs From Same Folder On Hierarchy Enter script**
    i. Moved this folder to the Runtime folder. This fixes the error where the engine can't find the script
    ii. Removed the code that made this script execute too early

- **Main Features Demo Scene**
    i. Changed name to "Showcase- RainWetness All Features"
    ii. Removed Vertex Paint on 1 duck. As it was masking out the rain effects
    iii. Add a few more ducks to showcase the Rain Occlusion effect

- **Rotating Object Demo**
    i. Changed name to "Showcase- RainWetness Shaders on Rotating Objects"

- **Default Demo**
    i. Changed name to "Showcase- RainWetness Shaders on Static Objects"

- **New Demos**
    i. Showcase- RainWetness Shaders- Random Colours Features

- **All Shader Demos**
    i. Changed names to "Showcase- RainWetness Shaders- (fill in shader name here)"

Version 1.0

- **Initial Version**

Version Alpha

- **DYNAMIC- RainWetness Prefab**
    i. This is the new home for all of the RainWetness scripts. Which helps reduce clutter on the player game object

- **Global Variables Script**
    i. Added this script to hold all shared variables for all of my scripts
        i. This helps increase performance and scalability

ii. Updated the script so that you can specify the player gameobject in the scene or you can allow the system to find the player automatically at the cost of start-up performance
iii. This script also controls the DO NOT DESTROY function
    i. This allows the RainWetness Systems to move between scenes

- **Rain Occlusion Zone Script**
    i. Renamed to Rain Occlusion Manager and make it so you only have to add this component to the player and not every rain occlusion trigger
    ii. Optimized Rain Occlusion Manager script to increase performance
    iii. Added a new script called Rain Occlusion Zone. Which communicates with the Rain Occlusion Manager script
        i. This increases performance exponentially

- **Interior Rain Audio Spawn Manager Script**
    i. Renamed to Dynamic Spawn Manager
        i. This will help with future updates and expanding of this scripts features

- **UI Navigation Script**
    i. Added function to saturate and desaturate UI when interacting with it
    ii. Updated entire script to work from 1 script instead of having it be on each button

- **Object Follower Script**
    i. Added an option to have an offset for the follow object (can be turned off with a toggle)

- **Editor UI**
    i. Add shortcuts to editor demos

- **All Shaders**
    i. Changed specular AA Screen Space Threshold and AA Threshold to 0.5. This fixes overly bright rain drops/streaks
    ii. Changed Rain Streak size, Rain Drop Size, and Rain Distortion Strength to global variables

- **Rain Sub-Shader**
    i. Changed the rain drop/streak UV/tiling to use the updated Biplaner UVs/ Biplaner UVs with time sub-shaders. This allows for all objects to have the same rain drop/streak tiling size
    ii. Changed rain streak texture sampler from regular to hex
    iii. Removed hex rotation variable
    iv. Added distortion normal strength variable
    v. Changed Rain Streak/Drops tiling size, and distortion strength to global variables, and added them to the Dynamic RainWetness Manager
    vi. Updated math for calculating rain streak normals
    vii. Cleaned up shader layout for better readability
    viii. Updated default settings for all variables
    ix. Replaced lots of nodes with sub-graphs for easier readability
    x. Removed some nodes in the Noise Distortion sub-shader

- **HDRP Shader- Iridescent/ Iridescent Tessellation**
    i. Added a bool to let you choose from using the objects tangents or a texture for the tangent map
    ii. Removediridescent mask bool
    iii. Removed iridescent mask min and max variables
    iv. Changed to 1 vector 2 remap
    v. Removed iridescent thickness bool
    vi. Removed iridescent thickness min and max variables
    vii. Changed to 1 vector 2 remap
    viii. Combined iridescent mask and iridescent thickness textures maps into 1
        1. R= Mask G= Thickness B= N/A
    ix. Fixed spelling errors
    x. Removed transparency settings

- **HDRP Shader- Anisotropy/ Anisotropy Tessellation**
    i. Changed the shaders name from Anisotropic to Anisotropy (as this is the correct name in Unity)
    ii. Removed transparent variables. As these aren't applicable to this shader

- **HDRP Shader- Transparent/ Transparent Tessellation**
    i. Changed the shaders name from Transparent to Translucent (as this is the correct name in Unity)

- **Biplaner UV Sub-Shaders**
  i. Changed name from “biplaner with time” to “Biplaner UVs- With Time”
  ii. Added outputs for tiling, offset, and uv’s
  iii. Removed the time nodes
  iv. Created a new sub-shader without the time nodes called “Biplaner UVs”

- **VFX Falling Rain (particle system)**
  i. Changed the shader to use the default soft particle shader.
    i. As the rain specific one had errors with changing the base textures color
  ii. Updated rain streaks to be more visible
  iii. Updated the base texture to have multiple rain streaks per particle
  iv. This increases performance and makes rains appear more dense
  v. Add a script that allows these particles to follow the player around the game world

- **Project Folders**
  i. Moved quite a lot of items to the new Shared folder
  ii. Organized and remained lots of items
  iii. Compressed some textures

- **Enviro Integration**
  i. Updated the Enviro Integration function in the Dynamic Rain Manager Script

- **Terrain RainWetness Decals**
  i. Added soft/hard prefab versions for all decals
  ii. Added/Removed/Updated exposed variables to increase ease of use
  iii. Renamed materials, shaders, and prefabs
  iv. Combined some nodes into subgraphs

- **MoveRotateResize First Time Setup Script**
  i. Combined script functions so they are all combined in the First Time Setup function. This lets the user setup my assest with one click
  ii. Updated/Cleaned up overall script functions and layout
  iii. Added turning on decals and decal layers
  iv. Removed NetCode Framework code
  v. Removed Allow Unsafe code

- **Change Shaders form A to B Script**
  i. Updated scripts functionality to let you bulk change more than one shader at a time
  ii. Updated GUI

- **Misc**
  i. Added the HierarchyDecorator add-on to clean up Unity’s Hierarchy interface (can be removed by deleting the folder in Third Party Assets)
  ii. Added the Naughty Attributes add-on to help organize Unity’s Interface for my scripts
  iii. Added definitions (example...#if HDRP #end) to multiple scripts to fix compiling errors
  iv. Added classes to all of my scripts to keep the code clean, organized, and to follow Unity’s store policy

- **Playable Demo**
  i. Added separate demo scenes for all shaders. Most demos includes a comparison model with the default shader
    1. Ducks = Rotating Objects
    2. Zombies = Skinned Mesh Renderers
    3. Statue = Default shader
    4. Horse = Anisotropic shader
    5. Seashell = Iridescent shader
    6. Potted Plant = Subsurface shader
    7. Glasses/Glass Ducks = Glass and Translucent shaders
  ii. Changed the player controller in all shader demo scenes (except the Occlusion Zone Demo) to my Dynamic: Fly Camera
  iii. Updated the main camera’s TAA sharpening settings
  iv. Updated the miscellaneous TAA settings in main camera

v. Updated some scenes post processing volume settings
vi. Fixed errors in some scenes post processing volume settings
vii. Added new demo scenes to showcase all my different shaders
viii. Added version# text to Main Menu UI
ix. Added F3 Toggle to some demos scenes which allow the object to rotate
x. Added F4 Toggle to some demos scenes which allow you to reset the objects rotation
xi. Added new script to control button UI navigation
xii. Added Sakai- SoftMask For UGUI (softens UI masks jagged edges)
xiii. Added info panels to all demo scenes
xiv. Updated Main Menu UI layout
xv. Updated all scenes logo size
xvi. Updated all scenes logo icons brightness
xvii. Updated Menu font sizes
xviii. Updated Demo Levels UI to Show F3 and F4
xix. Updated Graphics Settings UI and improved functionality

UI Main Menu Script

1. Updated pause function to work with the new Dynamic Fly Camera script

~~Version 1.3.5 (*NOT Released Yet)~~

~~★ !!STOP!! Process for updating to new version~~

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2. ~~Once that is completed please delete the entire MoveRotateResize folder that's inside of your Unity project~~
3. ~~Now you can update and install new version of the asset~~

- ~~**Rain Wetness Shaders**~~
  - i. ~~Added Default- Parallax Occlusion shader and sub-shaders~~
  - ii. ~~Updated all shaders Effects Mask to include a procedural wetmask option~~
  - iii. ~~Updated all shaders Effects Mask to allow coverage to be controlled via the global rain and global rain extended variables.~~

- ~~**Texture Painter**~~
  - i. ~~Look into adding GitHub - IRCSS/TexturePaint to allow users to paint the Effect Mask in the Unity editor~~

- ~~**DYNAMIC: Rain/Wetness window**~~
  - i. ~~Added new window that allows the user to easily access all the options and features of our asset~~

- ~~**DYNAMIC: Systems Manager**~~
  - i. ~~Added player collider and rigidbody variables~~

- ~~**Downloadable Demo**~~
  - i. ~~Added new environment for the users to walk around in (not included in editor demos)~~

- ~~**Editor Demos**~~
  - i. ~~Replaced Iridescent models with new CC0 ones~~
    1. ~~Oil Spill container - Download Free 3D model by Avadhoot (@avadhootnb11) [3b631c6]~~
    2. ~~Car Scene - Download Free 3D model by toivo~~
    3. ~~Leather Armor - Download Free 3D model by Blueprint Games (@Vahil) [0693c6c]~~
  - ii. ~~Replaced Anisotropic models with new CC0 ones~~
    1. ~~3DDecember 2021 - Copper Pots and Pans - Download Free 3D model by Jack Kelly (@SecretImbecile) [851f78f]~~
    2. ~~CD + Jewel Case - Download Free 3D model by blue man group (@bloobloo) [f6c8386]~~
  - iii. ~~Added demo to showcase Parallax Occlusion shader~~